

## 48. SD Host Interface (SDHI)

This MCU incorporates an SD host interface (SDHI) which is compliant with the SD Specifications. When developing host devices that are compliant with the SD Specifications, the user must enter into the SD Host/Ancillary Product License Agreement (SD HALA).

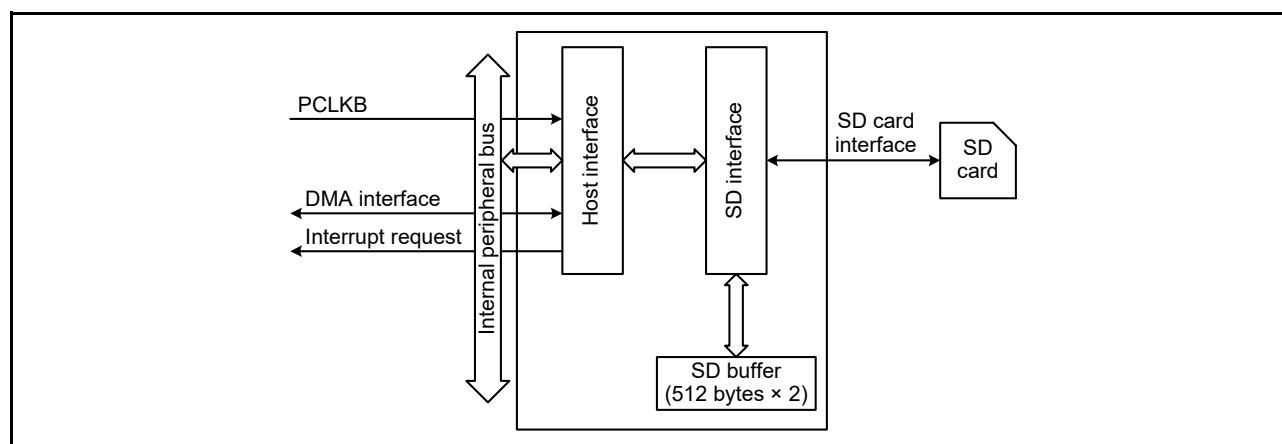
### 48.1 Overview

Table 48.1 lists the SDHI specifications.

**Table 48.1 SDHI Specifications**

| Item                  | Description                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SD bus interface      | <ul style="list-style-type: none"> <li>Compatible with SD memory card and SDIO card (<b>NOT</b> compatible with the SPI bus interface, embedded SDIO shared bus, 8-bit SD bus, or SDIO suspend/resume functions)</li> <li>Transfer bus mode selectable from 4-bit wide bus mode or 1-bit default bus mode</li> <li>Compatible with SD, SDHC, and SDXC formats</li> </ul> |
| Transfer modes        | Selectable from high-speed mode or default speed mode                                                                                                                                                                                                                                                                                                                    |
| SDHI clock            | The SDHI clock is generated by dividing peripheral module clock B (PCLKB) by $n$ , where $n = 1, 2, 4, 8, 16, 32, 64, 128, 256$ , or 512                                                                                                                                                                                                                                 |
| Error check functions | <ul style="list-style-type: none"> <li>CRC7 (command/response)</li> <li>CRC16 (transfer data)</li> </ul>                                                                                                                                                                                                                                                                 |
| Interrupt sources     | <ul style="list-style-type: none"> <li>Card access interrupt (CAI)</li> <li>SDIO access interrupt (SDACI)</li> <li>Card detection interrupt (CDETI)</li> <li>SD buffer access interrupt (SBFAI)</li> </ul>                                                                                                                                                               |
| DMA transfer sources  | <ul style="list-style-type: none"> <li>DMAC and DTC triggerable by the SBFAI interrupt</li> <li>SD buffer is read and write accessible using the DMAC and DTC</li> </ul>                                                                                                                                                                                                 |
| Other functions       | <ul style="list-style-type: none"> <li>Card detection</li> <li>Write protection</li> </ul>                                                                                                                                                                                                                                                                               |

Figure 48.1 shows a block diagram of the SDHI.



**Figure 48.1 Block Diagram of the SDHI**

Table 48.2 lists the pin configuration of the SDHI.

**Table 48.2 Pin Configuration of the SDHI**

| Pin Name | I/O    | Description                          |
|----------|--------|--------------------------------------|
| SDHI_CLK | Output | SDHI clock                           |
| SDHI_CMD | I/O    | Command output, response input       |
| SDHI_D0  | I/O    | Data 0 (DAT0)                        |
| SDHI_D1  | I/O    | Data 1 (DAT1), SDIO access interrupt |
| SDHI_D2  | I/O    | Data 2 (DAT2), read wait             |
| SDHI_D3  | I/O    | Data 3 (DAT3), SD card detection     |
| SDHI_CD  | Input  | SD card detection                    |
| SDHI_WP  | Input  | SD card write protection             |

## 48.2 Register Details

### 48.2.1 Command Register (SDCMD)

Address(es): SDHI.SDCMD 0008 AC00h

|                    |       |       |       |            |     |     |           |     |             |     |     |     |     |     |     |
|--------------------|-------|-------|-------|------------|-----|-----|-----------|-----|-------------|-----|-----|-----|-----|-----|-----|
| b31                | b30   | b29   | b28   | b27        | b26 | b25 | b24       | b23 | b22         | b21 | b20 | b19 | b18 | b17 | b16 |
| —                  | —     | —     | —     | —          | —   | —   | —         | —   | —           | —   | —   | —   | —   | —   | —   |
| Value after reset: | 0     | 0     | 0     | 0          | 0   | 0   | 0         | 0   | 0           | 0   | 0   | 0   | 0   | 0   | 0   |
| b15                | b14   | b13   | b12   | b11        | b10 | b9  | b8        | b7  | b6          | b5  | b4  | b3  | b2  | b1  | b0  |
| CMD12AT[1:0]       | TRSTP | CMDRW | CMDTP | RSPTP[2:0] |     |     | ACMD[1:0] |     | CMDIDX[5:0] |     |     |     |     |     |     |
| Value after reset: | 0     | 0     | 0     | 0          | 0   | 0   | 0         | 0   | 0           | 0   | 0   | 0   | 0   | 0   | 0   |

| Bit        | Symbol       | Bit Name                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | R/W |
|------------|--------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b5 to b0   | CMDIDX[5:0]  | Command Index Field Value Select  | These bits configure the command index field value. The examples below include the bit values for the ACMD[1:0] bits.<br><div style="display: flex; justify-content: space-between;"> <span>b7</span> <span>b0</span> </div> 0 0 0 0 1 1 0: CMD6<br>0 0 0 1 0 0 1 0: CMD18<br>0 1 0 0 1 1 0 1: ACMD13                                                                                                                                                                                                                                                                                                   | R/W |
| b7, b6     | ACMD[1:0]    | Command Type Select               | <div style="display: flex; justify-content: space-between;"> <span>b7 b6</span> </div> 0 0: CMD<br>0 1: ACMD<br>Only set the values listed above.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W |
| b10 to b8  | RSPTP[2:0]   | Response Type Select *1           | <div style="display: flex; justify-content: space-between;"> <span>b10</span> <span>b8</span> </div> 0 0 0: Normal mode. Depending on the command, the response type and transfer method are selected by setting the ACMD[1:0] bits and CMDIDX[5:0] bits. At this time, the values for b15 to b11 in this register are invalid.<br>0 1 1: Expansion mode and no response<br>1 0 0: Expansion mode and R1, R5, R6, or R7 response<br>1 0 1: Expansion mode and R1b response<br>1 1 0: Expansion mode and R2 response<br>1 1 1: Expansion mode and R3 or R4 response<br>Only set the values listed above. | R/W |
| b11        | CMDTP        | Data Transfer Select *2           | 0: Command does not include data transfer (bc, bcr, or ac)<br>1: Command includes data transfer (adtc)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W |
| b12        | CMDRW        | Data Transfer Direction Select *3 | 0: Write data to the SD card<br>1: Read data from the SD card                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | R/W |
| b13        | TRSTP        | Block Transfer Select *3          | 0: Single block transferred<br>1: Multiple blocks transferred                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | R/W |
| b15, b14   | CMD12AT[1:0] | CMD12 Automatic Issue Select *4   | <div style="display: flex; justify-content: space-between;"> <span>b15</span> <span>b14</span> </div> 0 0: CMD12 is automatically issued during multi-block transfer<br>0 1: CMD12 is not automatically issued during multi-block transfer<br>Only set the values listed above.                                                                                                                                                                                                                                                                                                                         | R/W |
| b31 to b16 | —            | Reserved                          | These bits are 0 when read and cannot be modified.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | R   |

Note 1. Some commands cannot be used in normal mode. Refer to Table 48.3 and set the RSPTP[2:0] bits.

Note 2. The CMDTP bit is valid only when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b.

Note 3. Bits CMDRW and TRSTP are valid only when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b, and the CMDTP bit is 1.

Note 4. The CMD12AT[1:0] bits are valid only when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b, and the TRSTP bit is 1.

The command type and response type are set in the SDCMD register. The command type and transfer mode must be set when the RSPTP[2:0] bits are 011b, 100b, 101b, 110b, or 111b. The sequence starts when a value is written to this register. Refer to Table 48.3 for setting examples. Do not write to the SDCMD register when the SDSTS2.CBSY flag is 1.

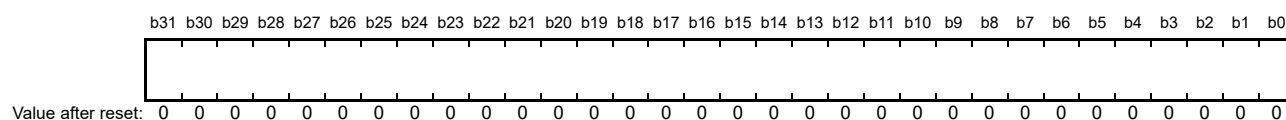
Table 48.3 lists examples of SDCMD register settings.

**Table 48.3 Examples of SDCMD Register Settings**

| Type | Command Symbol | SDCMD Register Setting   | Remarks                                                                                                                                                                                              |
|------|----------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMD  | CMD0           | 0000 0000h               |                                                                                                                                                                                                      |
|      | CMD2           | 0000 0002h               |                                                                                                                                                                                                      |
|      | CMD3           | 0000 0003h               |                                                                                                                                                                                                      |
|      | CMD4           | 0000 0004h               |                                                                                                                                                                                                      |
|      | CMD5           | 0000 0705h or 0000 0005h |                                                                                                                                                                                                      |
|      | CMD6           | 0000 1C06h or 0000 0006h |                                                                                                                                                                                                      |
|      | CMD7           | 0000 0007h               | When the card is deselected, the SD card does not return a response, so the SDSTS2.RSPTO flag becomes 1.                                                                                             |
|      | CMD8           | 0000 0408h or 0000 0008h |                                                                                                                                                                                                      |
|      | CMD9           | 0000 0009h               |                                                                                                                                                                                                      |
|      | CMD10          | 0000 000Ah               |                                                                                                                                                                                                      |
|      | CMD11          | 0000 040Bh or 0000 000Bh |                                                                                                                                                                                                      |
|      | CMD12          | 0000 000Ch               |                                                                                                                                                                                                      |
|      | CMD13          | 0000 000Dh               |                                                                                                                                                                                                      |
|      | CMD15          | 0000 000Fh               |                                                                                                                                                                                                      |
|      | CMD16          | 0000 0010h               |                                                                                                                                                                                                      |
|      | CMD17          | 0000 0011h               |                                                                                                                                                                                                      |
|      | CMD18          | 0000 0012h               |                                                                                                                                                                                                      |
|      | CMD20          | 0000 0514h or 0000 0014h |                                                                                                                                                                                                      |
|      | CMD24          | 0000 0018h               |                                                                                                                                                                                                      |
|      | CMD25          | 0000 0019h               |                                                                                                                                                                                                      |
|      | CMD27          | 0000 001Bh               |                                                                                                                                                                                                      |
|      | CMD28          | 0000 001Ch               |                                                                                                                                                                                                      |
|      | CMD29          | 0000 001Dh               |                                                                                                                                                                                                      |
|      | CMD30          | 0000 001Eh               |                                                                                                                                                                                                      |
|      | CMD32          | 0000 0020h               |                                                                                                                                                                                                      |
|      | CMD33          | 0000 0021h               |                                                                                                                                                                                                      |
|      | CMD38          | 0000 0026h               |                                                                                                                                                                                                      |
|      | CMD42          | 0000 002Ah               |                                                                                                                                                                                                      |
|      | CMD52          | 0000 0434h or 0000 0034h |                                                                                                                                                                                                      |
|      | CMD53          | 0000 1C35h               | Single block read                                                                                                                                                                                    |
|      |                | 0000 0C35h               | Single block write                                                                                                                                                                                   |
|      |                | 0000 7C35h               | Multi-block read                                                                                                                                                                                     |
|      |                | 0000 6C35h               | Multi-block write                                                                                                                                                                                    |
|      |                | 0000 0035h               | The setting to the left can be used regardless of the transfer being single block or multi-block. However, the MSB in the SDARG register (RW flag) must be set to 0 when reading and 1 when writing. |
|      | CMD55          | 0000 0037h               |                                                                                                                                                                                                      |
|      | CMD56          | 0000 0038h               |                                                                                                                                                                                                      |
| ACMD | ACMD6          | 0000 0046h               |                                                                                                                                                                                                      |
|      | ACMD13         | 0000 004Dh               |                                                                                                                                                                                                      |
|      | ACMD22         | 0000 0056h               |                                                                                                                                                                                                      |
|      | ACMD23         | 0000 0057h               |                                                                                                                                                                                                      |
|      | ACMD41         | 0000 0069h               |                                                                                                                                                                                                      |
|      | ACMD42         | 0000 006Ah               |                                                                                                                                                                                                      |
|      | ACMD51         | 0000 0073h               |                                                                                                                                                                                                      |

### 48.2.2 Argument Register (SDARG)

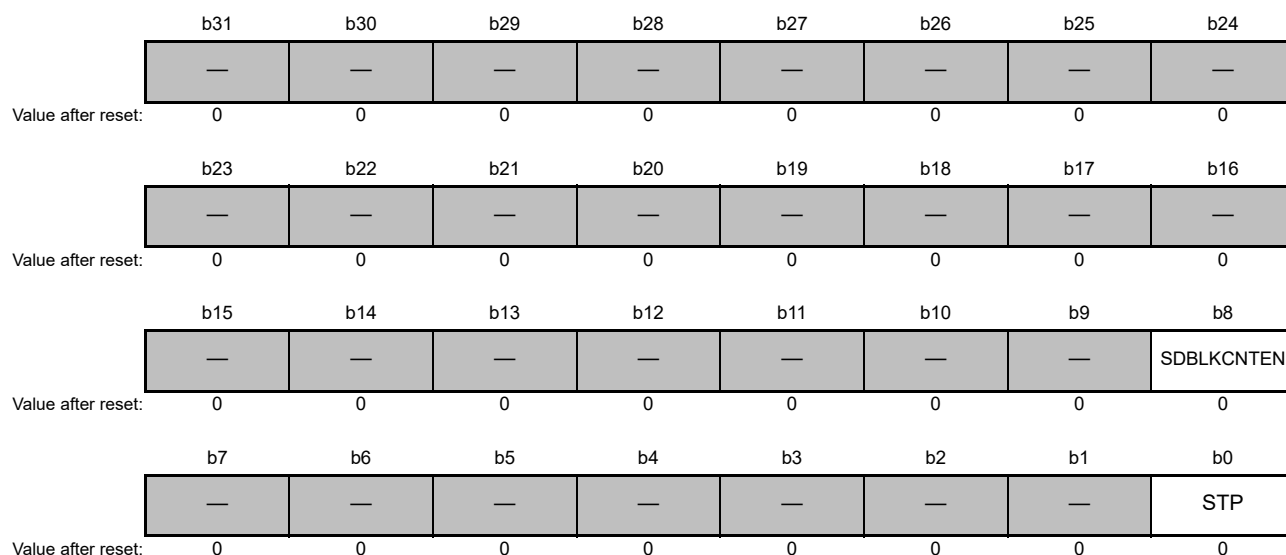
Address(es): SDHI.SDARG 0008 AC08h



The SDARG register is used for setting the argument field value. Set the SDARG register before setting the SDCMD register. The argument field value of the automatically issued CMD12 is 0000 0000h regardless of the SDARG register value.

### 48.2.3 Data Stop Register (SDSTOP)

Address(es): SDSTOP 0008 AC10h



| Bit       | Symbol      | Bit Name                             | Description                                                                  | R/W |
|-----------|-------------|--------------------------------------|------------------------------------------------------------------------------|-----|
| b0        | STP         | Transfer Stop                        | Data transfer stops when this bit is set to 1.                               | R/W |
| b7 to b1  | —           | Reserved                             | These bits are 0 when read and cannot be modified.                           | R   |
| b8        | SDBLKCN TEN | Block Count Register Value Select *1 | 0: SDBLKCNT register value is invalid<br>1: SDBLKCNT register value is valid | R/W |
| b31 to b9 | —           | Reserved                             | These bits are 0 when read and cannot be modified.                           | R   |

Note 1. Do not rewrite this bit when the SDSTS2.CBSY flag is 1.

The SDSTOP register stops data transfer. During a multi-block transfer sequence, the SDBLKCNT register value (number of blocks to be transferred) can be set to valid or invalid by setting the SDSTOP register.

#### STP Bit (Transfer Stop)

When setting the STP bit to 1, set it after the SDSTS1.RSPEND flag becomes 1; when setting the STP bit to 0, set it after the SDSTS1.ACEND flag becomes 1. After a command sequence is complete, the SDHI does not issue CMD12 and the SDSTS1.ACEND flag does not become 1 even if the STP bit is set to 1. When the SDHI is in the busy state after receiving the R1b response, the SDHI does not issue CMD12 even if the STP bit is 1, and after the SDHI is released from

the busy state, the SDSTS1.ACEND flag becomes 1.

- Performing a multi-block transfer

When the STP bit is set to 1, the SDHI issues CMD12, and the command sequence is stopped. The SD buffer can be accessed even after the STP bit is set to 1, but a buffer access error occurs and the SDSTS2.ILW flag or SDSTS2.ILR flag becomes 1. If the command sequence stops due to a communication error or a timeout, the SDHI does not issue CMD12.

- Performing a single block transfer

When the STP bit is set to 1 during a single block write access, if there is no data in the SD buffer, the SDSTS1.ACEND flag becomes 1. If there is data in the SD buffer, after the SDHI is released from the busy state, the SDSTS1.ACEND flag becomes 1. When the STP bit is set to 1 during a single block read access, the SDSTS1.ACEND flag becomes 1. Also, CMD12 is not issued even if the STP bit is set to 1 during a single block read access or single block write access.

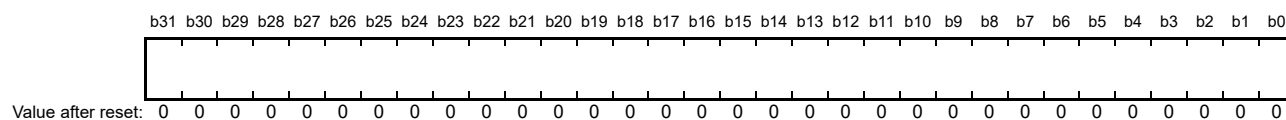
### SDBLKCNTEN Bit (Block Count Register Value Select)

If the SDBLKCNTEN bit is 1 during a multi-block transfer sequence, the SDHI automatically issues CMD12. When the SDCMD.RSPTP[2:0] bits are set to 000b and CMD18 or CMD25 is issued, or if the SDCMD.RSPTP[2:0] bits are set to 011b, 100b, 101b, 110b, or 111b and the SDCMD.TRSTP bit is 1 (multiple blocks transferred), if the SDCMD.CMD12AT[1:0] bits are 00b (CMD12 is automatically issued during multi-block transfer), and the number of transfer blocks reaches the value set in the SDBLKCNT register, the SDHI automatically issues CMD12.

If the command sequence is stopped by a communication error or a timeout, CMD12 is not automatically issued.

## 48.2.4 Block Count Register (SDBLKCNT)

Address(es): SDHI.SDBLKCNT 0008 AC14h

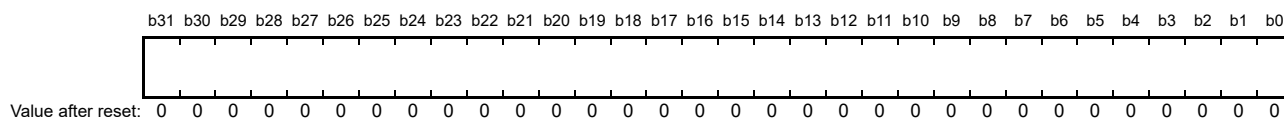


When performing a multi-block transfer, SDBLKCNT is a readable/writable register used to set the number of blocks to be transferred. For example, when the register value is 0000 0001h, 1 block is transferred; when the register value is 0000 FFFFh, 65,535 blocks are transferred; and when the register value is FFFF FFFFh, 4,294,967,295 blocks are transferred. Do not set this register to 0000 0000h. Do not rewrite the SDBLKCNT register when the SDSTS2.CBSY flag is 1.

### 48.2.5 Response Register 10 (SDRSP10), Response Register 32 (SDRSP32), Response Register 54 (SDRSP54), Response Register 76 (SDRSP76)

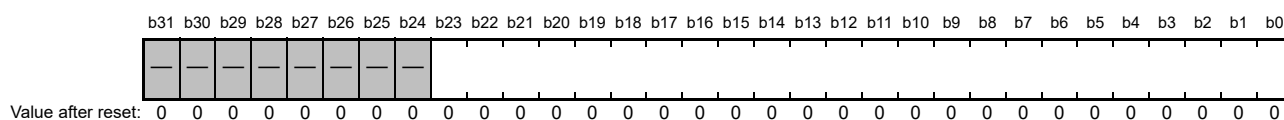
- SDRSP10, SDRSP32, SDRSP54

Address(es): SDHI.SDRSP10 0008 AC18h, SDHI.SDRSP32 0008 AC20h, SDHI.SDRSP54 0008 AC28h



- SDRSP76

Address(es): SDHI.SDRSP76 0008 AC30h



| Bit        | Symbol | Bit Name | Description                                         | R/W |
|------------|--------|----------|-----------------------------------------------------|-----|
| b23 to b0  | —      | —        | This register stores the response from the SD card. | R   |
| b31 to b24 | —      | Reserved | These bits are 0 when read.                         | R   |

Registers SDRSP10, SDRSP32, SDRSP54, and SDRSP76 are read-only registers that store the response from the SD card. Depending on the type of response from the SD card, the SDHI divides and stores the response among the four registers.

Table 48.4 lists the correspondence between the response type and its storage destination.

**Table 48.4 Correspondence Between the Response Type and Its Storage Destination**

| Response Type | SDRSP10 Register | SDRSP32 Register | SDRSP54 Register | SDRSP76 Register |
|---------------|------------------|------------------|------------------|------------------|
| R1            | [39:8]           | —                | [39:8] *1        | —                |
| R1b           | [39:8]           | —                | [39:8] *1        | —                |
| R2            | [39:8]           | [71:40]          | [103:72]         | [127:104]        |
| R3            | [39:8]           | —                | —                | —                |
| R4            | [39:8]           | —                | —                | —                |
| R5            | [39:8]           | —                | —                | —                |
| R6            | [39:8]           | —                | —                | —                |
| R7            | [39:8]           | —                | —                | —                |

Note 1. The response for CMD18 and CMD25 is stored in registers SDRSP10 and SDRSP54. Therefore, even if the SDRSP10 register is overwritten with the response for the automatically transmitted CMD12, the response for CMD18 or CMD25 can be confirmed by reading the SDRSP54 register.

## 48.2.6 SD Status Register 1 (SDSTS1)

Address(es): SDHI.SDSTS1 0008 AC38h

|                    |     |         |        |        |         |        |        |
|--------------------|-----|---------|--------|--------|---------|--------|--------|
| b31                | b30 | b29     | b28    | b27    | b26     | b25    | b24    |
| —                  | —   | —       | —      | —      | —       | —      | —      |
| Value after reset: | 0   | 0       | 0      | 0      | 0       | 0      | 0      |
| b23                | b22 | b21     | b20    | b19    | b18     | b17    | b16    |
| —                  | —   | —       | —      | —      | —       | —      | —      |
| Value after reset: | 0   | 0       | 0      | 0      | 0       | 0      | 0      |
| b15                | b14 | b13     | b12    | b11    | b10     | b9     | b8     |
| —                  | —   | —       | —      | —      | SDD3MON | SDD3IN | SDD3RM |
| Value after reset: | 0   | 0       | 0      | 0      | x       | 0      | 0      |
| b7                 | b6  | b5      | b4     | b3     | b2      | b1     | b0     |
| SDWPMON            | —   | SDCDMON | SDCDIN | SDCDRM | ACEND   | —      | RSPEND |
| Value after reset: | x   | 0       | x      | 0      | 0       | 0      | 0      |

| Bit        | Symbol  | Bit Name                    | Description                                                                                              | R/W      |
|------------|---------|-----------------------------|----------------------------------------------------------------------------------------------------------|----------|
| b0         | RSPEND  | Response End Detection Flag | 0: Response end is not detected<br>1: Response end is detected                                           | R/(W) *1 |
| b1         | —       | Reserved                    | This bit is 0 when read and cannot be modified.                                                          | R        |
| b2         | ACEND   | Access End Detection Flag   | 0: Access end is not detected<br>1: Access end is detected                                               | R/(W) *1 |
| b3         | SDCDRM  | SDHI_CD Removal Flag        | 0: SD card removal not detected by the SDHI_CD pin<br>1: SD card removal detected by the SDHI_CD pin     | R/(W) *1 |
| b4         | SDCDIN  | SDHI_CD Insertion Flag      | 0: SD card insertion not detected by the SDHI_CD pin<br>1: SD card insertion detected by the SDHI_CD pin | R/(W) *1 |
| b5         | SDCDMON | SDHI_CD Pin Monitor Flag    | 0: SDHI_CD pin level is high *2<br>1: SDHI_CD pin level is low *2                                        | R        |
| b6         | —       | Reserved                    | This bit is 0 when read and cannot be modified.                                                          | R        |
| b7         | SDWPMON | SDHI_WP Pin Monitor Flag    | 0: SDHI_WP pin level is high<br>1: SDHI_WP pin level is low                                              | R        |
| b8         | SDD3RM  | SDHI_D3 Removal Flag        | 0: SD card removal not detected by the SDHI_D3 pin<br>1: SD card removal detected by the SDHI_D3 pin     | R/(W) *1 |
| b9         | SDD3IN  | SDHI_D3 Insertion Flag      | 0: SD card insertion not detected by the SDHI_D3 pin<br>1: SD card insertion detected by the SDHI_D3 pin | R/(W) *1 |
| b10        | SDD3MON | SDHI_D3 Pin Monitor Flag    | 0: SDHI_D3 pin level is low<br>1: SDHI_D3 pin level is high                                              | R        |
| b31 to b11 | —       | Reserved                    | These bits are 0 when read and cannot be modified.                                                       | R        |

Note 1. The flag does not change even if set to 1. Writing 0 changes the flag value to 0.

Note 2. The flag changes when the pin level continues for the period set in the SDOPT.CTOP[3:0] bits or longer.

The SDSTS1 register indicates the detection of a response end or access end for a command sequence. The SDSTS1 register also indicates the detection SD card insertion/removal, and indicates the write protection status.

During a multi-block transfer sequence, if CMD12 or CMD52 (SDIO abort) is issued, the ACEND flag becomes 1, but the RSPEND flag remains set to 0.

If the command sequence is stopped due to a communication error or timeout, the ACEND flag or RSPEND flag becomes 1.

After a reset is released, the SDD3MON flag, SDD3IN flag, and SDD3RM flag values are changed according to the status of the SDHI\_D3 pin, and their values are changed when data is being transferred in wide bus mode.

Flags to be cleared should be set to 0; flags not being cleared should be set to 1.

**RSPEND Flag (Response End Detection Flag)**

— This flag becomes 1 under any of the following conditions:

- A response is received.
- A command that does not have a response is issued.
- After the R1b response is received, the SDHI is released from the busy state.
- During a multi-block transmission, after the SDIOMD.C52PUB bit is set to 1, the CMD52 response is received.
- A communication error or timeout causes the command sequence to abort.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

Note: When a command is issued that is absent of data transfer, the RSPEND flag becomes 1 after the command sequence ends.

**ACEND Flag (Access End Detection Flag)**

— This flag becomes 1 under any of the following conditions:

- During a single block read sequence, the SD buffer read access is completed.
- During a multi-block read sequence, the last block is read from the SD buffer.
- During a multi-block read sequence, if CMD12 is automatically issued, data is read from the SD buffer, and the response for CMD12 is received.
- During a single block write sequence, after a CRC status token is received, the SDHI is released from the busy state.
- During a multi-block write sequence, after a CRC status token is received for the last block, the SDHI is released from the busy state.
- During a multi-block write sequence, when CMD12 is automatically issued, a response busy of the automatically issued CMD12 is received.
- During a multi-block read sequence, when CMD12 is automatically issued, after setting the SDSTOP.STP bit to 1, a response of the automatically issued CMD12 is received.
- During a multi-block write sequence, when CMD12 is automatically issued, after setting the SDSTOP.STP bit to 1, a response busy of the automatically issued CMD12 is received.
- During a multi-block read sequence, after the SDIOMD.IOABT bit is set to 1, the response for CMD52 is received.
- During a multi-block write sequence, after the SDIOMD.IOABT bit is set to 1, the response for CMD52 is received.
- A communication error or timeout causes the command sequence to abort.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

Note: The ACEND flag becomes 1 after the command sequence ends.

**SDCDRM Flag (SDHI\_CD Removal Flag)**

— This flag becomes 1 under the following condition:

- The SDHI\_CD pin changes from low to high, and the high period is the period set in the SDOPT.CTOP[3:0] bits or longer.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

**SDCDIN Flag (SDHI\_CD Insertion Flag)**

— This flag becomes 1 under the following condition:

- The SDHI\_CD pin changes from high to low, and the low period is the period specified in the SDOPT.CTOP[3:0] bits or longer.

— This flag becomes 0 under the following condition:

- The flag is set to 0.



**SDD3RM Flag (SDHI\_D3 Removal Flag)**

— This flag becomes 1 under the following condition:

- The SDHI\_D3 pin changes from high to low, and the low period is at least two PCLKB cycles.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

**SDD3IN Flag (SDHI\_D3 Insertion Flag)**

— This flag becomes 1 under the following condition:

- The SDHI\_D3 pin changes from low to high, and the high period is at least two PCLKB cycles.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

## 48.2.7 SD Status Register 2 (SDSTS2)

Address(es): SDHI.SDSTS2 0008 AC3Ch

|                    |       |           |     |     |      |      |      |
|--------------------|-------|-----------|-----|-----|------|------|------|
| b31                | b30   | b29       | b28 | b27 | b26  | b25  | b24  |
| —                  | —     | —         | —   | —   | —    | —    | —    |
| Value after reset: | 0     | 0         | 0   | 0   | 0    | 0    | 0    |
| b23                | b22   | b21       | b20 | b19 | b18  | b17  | b16  |
| —                  | —     | —         | —   | —   | —    | —    | —    |
| Value after reset: | 0     | 0         | 0   | 0   | 0    | 0    | 0    |
| b15                | b14   | b13       | b12 | b11 | b10  | b9   | b8   |
| ILA                | CBSY  | SDCLKCREN | —   | —   | —    | BWE  | BRE  |
| Value after reset: | 0     | 0         | 1   | 0   | 0    | 0    | 0    |
| b7                 | b6    | b5        | b4  | b3  | b2   | b1   | b0   |
| SDD0MON            | RSPTO | ILR       | ILW | DTO | ENDE | CRCE | CMDE |
| Value after reset: | x     | 0         | 0   | 0   | 0    | 0    | 0    |

| Bit        | Symbol    | Bit Name                                   | Description                                                                                                                                                                                                                               | R/W      |
|------------|-----------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| b0         | CMDE      | Command Error Detection Flag               | 0: Command error not detected<br>1: Command error detected                                                                                                                                                                                | R/(W) *1 |
| b1         | CRCE      | CRC Error Detection Flag                   | 0: CRC error not detected<br>1: CRC error detected                                                                                                                                                                                        | R/(W) *1 |
| b2         | ENDE      | End Bit Error Detection Flag               | 0: End bit error not detected<br>1: End bit error detected                                                                                                                                                                                | R/(W) *1 |
| b3         | DTO       | Data Timeout Detection Flag                | 0: Data timeout not detected<br>1: Data timeout detected                                                                                                                                                                                  | R/(W) *1 |
| b4         | ILW       | SDBUFR Illegal Write Access Detection Flag | 0: Illegal write access to the SDBUFR register not detected<br>1: Illegal write access to the SDBUFR register detected                                                                                                                    | R/(W) *1 |
| b5         | ILR       | SDBUFR Illegal Read Access Detection Flag  | 0: Illegal read access to the SDBUFR register not detected<br>1: Illegal read access to the SDBUFR register detected                                                                                                                      | R/(W) *1 |
| b6         | RSPTO     | Response Timeout Detection Flag            | 0: Response timeout not detected<br>1: Response timeout detected                                                                                                                                                                          | R/(W) *1 |
| b7         | SDD0MON   | SDHI_D0 Pin Status Flag                    | 0: SDHI_D0 pin is low<br>1: SDHI_D0 pin is high                                                                                                                                                                                           | R        |
| b8         | BRE       | SDBUFR Read Enable Flag                    | 0: Read access to the SDBUFR register disabled<br>1: Read access to the SDBUFR register enabled                                                                                                                                           | R/(W) *1 |
| b9         | BWE       | SDBUFR Write Enable Flag                   | 0: Write access to the SDBUFR register disabled<br>1: Write access to the SDBUFR register enabled                                                                                                                                         | R/(W) *1 |
| b10        | —         | Reserved                                   | This bit is 0 when read and cannot be modified.                                                                                                                                                                                           | R        |
| b11        | —         | Reserved                                   | This bit is 0 when read. Set it to 1 when writing.                                                                                                                                                                                        | R/W      |
| b12        | —         | Reserved                                   | This bit is 0 when read and cannot be modified.                                                                                                                                                                                           | R        |
| b13        | SDCLKCREN | SDCLKCR Write Enable Flag                  | 0: SD bus (CMD and DAT lines) is busy, so write access to the SDCLKCR.CLKEN bit and CLKSEL[7:0] bits is disabled.<br>1: SD bus (CMD and DAT lines) is not busy, so write access to the SDCLKCR.CLKEN bit and CLKSEL[7:0] bits is enabled. | R        |
| b14        | CBSY      | Command Sequence Status Flag               | 0: Command sequence completed<br>1: Command sequence in progress (busy)                                                                                                                                                                   | R        |
| b15        | ILA       | Illegal Access Error Detection Flag        | 0: Illegal access error not detected<br>1: Illegal access error detected                                                                                                                                                                  | R/(W) *1 |
| b31 to b16 | —         | Reserved                                   | These bits are 0 when read and cannot be modified.                                                                                                                                                                                        | R        |

Note 1. The flag does not change even if set to 1. Writing 0 changes the flag value to 0.

The SDSTS2 register indicates the status of the SD buffer and the status of the SD card. Flags to be cleared should be set to 0; flags not being cleared should be set to 1.

#### CMDE Flag (Command Error Detection Flag)

The command sequence is stopped when a command error occurs. When the SDIOMD.C52PUB bit is set to 1 and CMD52 is automatically issued, if a communication error or response timeout occurs, the command sequence will not be completed. Perform the error processing shown in section 48.3.6.8 or section 48.3.6.9 and complete the command sequence.

— This flag becomes 1 under any of the following conditions:

- The command index field value for the command transmitted differs from the command index field value for the response received.
- The command index field value for the automatically issued CMD12 or CMD52 (which are the commands to stop transfer) differs from the command index field value for the response received.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

#### CRCE Flag (CRC Error Detection Flag)

The command sequence is stopped when a CRC error occurs. When the SDIOMD.C52PUB bit is set to 1 and CMD52 is automatically issued, if a communication error or response timeout occurs, the command sequence will not be completed. Perform the error processing shown in section 48.3.6.8 or section 48.3.6.9 and complete the command sequence.

— This flag becomes 1 under any of the following conditions:

- The received CRC status token is in error (the value of the CRC status is a value other than 010b).
- The read data contains a CRC error.
- The response contains a CRC error.
- The response for the automatically issued CMD12 or CMD52 (which are the commands to stop transfer) contains a CRC error.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

#### ENDE Flag (End Bit Error Detection Flag)

The command sequence is stopped when an end bit error occurs. When the SDIOMD.C52PUB bit is set to 1 and CMD52 is automatically issued, if a communication error or response timeout occurs, the command sequence will not be completed. Perform the error processing shown in section 48.3.6.8 or section 48.3.6.9 and complete the command sequence.

— This flag becomes 1 under any of the following conditions:

- The response length is in error (the end bit could not be detected).
- The read data length is in error (the end bit of the enabled bit could not be detected).
- The CRC status token length is in error (the end bit could not be detected).
- The response length for the automatically issued CMD12 or CMD52 (which are the commands to stop transfer) contains an error (the end bit could not be detected).

— This flag becomes 0 under the following condition:

- The flag is set to 0.

**DTO Flag (Data Timeout Detection Flag)**

This flag indicates that the data expected to be received during the period specified (set in the SDOPT.TOP[3:0] bits) was not received. However, response timeouts are excluded. The command sequence stops when a data timeout occurs.

— This flag becomes 1 under any of the following conditions:

- After the R1b response is received, the SDHI is busy for the period specified or longer.
- After the CRC status token is received, the SDHI is busy for the period specified or longer.
- After data is written, the CRC status token is not received even after the period specified elapsed.
- After a read command is issued, the read data is not received even after the period specified elapsed.
- After CMD12 is issued during a command sequence, the SDHI is busy for the period specified or longer.
- After the read data is received, the next read data is not received even after the period specified elapsed.
- After the SDHI exits the read wait state, the next read data is not received even after the period specified elapsed.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

**ILW Flag (SDBUFR Illegal Write Access Detection Flag)**

— This flag becomes 1 under any of the following conditions:

- A value is written to the SDBUFR register while the SDHI is not in the data read or data write command state.
- A value is written to the SDBUFR register while the SD buffer is full.
- A value is written to the SDBUFR register while the CRC status token or CRC status token length is in error.
- After the CRC status token is received, a value is written to the SDBUFR register if the SDHI is busy for the period set in bits SDOPT.TOP[3:0] or longer.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

**ILR Flag (SDBUFR Illegal Read Access Detection Flag)**

— This flag becomes 1 under any of the following conditions:

- The SDBUFR register is read while the SD buffer is empty.
- The value read from the SDBUFR register includes a CRC error or end bit error.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

**RSPTO Flag (Response Timeout Detection Flag)**

The command sequence is stopped when a response timeout occurs. When the SDIOMD.C52PUB bit is set to 1 and CMD52 is automatically issued, if a communication error or response timeout occurs, the command sequence will not be completed. Perform the error processing shown in section 48.3.6.8 or section 48.3.6.9 and complete the command sequence.

— This flag becomes 1 under the following condition:

- A response is not received even after 640 SDHI clock cycles or more have elapsed (including the response for the automatically issued CMD12 or CMD52 (which are the commands to stop transfer)).

— This flag becomes 0 under the following condition:

- The flag is set to 0.

**SDD0MON Flag (SDHI\_D0 Pin Status Flag)**

This flag indicates the status of the SDHI\_D0 pin. After an erase command is issued, if the DTO flag is 1 and the RSPTO flag is 0, polling can be used to monitor the SDD0MON flag change from 0 to 1, and check that the erase command sequence is complete. If a communication error or timeout occurs during the write sequence, the SDHI\_D0 pin may remain low. When the SDHI clock is stopped, the value before the SDHI clock was stopped is retained.

**BRE Flag (SDBUFR Read Enable Flag)**

— This flag becomes 1 under any of the following conditions:

- During a single block transfer, the data size set in the SDSIZE.LEN[9:0] bits is stored in the SD buffer.
- During a multi-block transfer, the data size set in the SDSIZE.LEN[9:0] bits is stored in one of the two SD buffers.

— This flag becomes 0 under any of the following conditions:

- The bit is set to 0.
- DMA transfer is used to read 1 block of data from the SD buffer.

If the CPU is used to read data from the SDBUFR register, set the BRE flag to 0 before reading the data size <sup>\*1</sup> set in the SDSIZE.LEN[9:0] bits. Even if the block of data read contains a CRC error or end bit error, the data is stored in the SD buffer and the BRE flag becomes 1.

Note 1. If the transfer data size set in the SDSIZE.LEN[9:0] bits is an odd number, the odd numbered byte is ignored. Refer to section 48.5.2, SDBUFR Register Illegal Write Error for details.

**BWE Flag (SDBUFR Write Enable Flag)**

— This flag becomes 1 under any of the following conditions:

- During a single block transfer, the SD buffer is empty.
- During a multi-block transfer, bank 1 or bank 2 of the SD buffer is empty.

— This flag becomes 0 under any of the following conditions:

- The flag is set to 0.
- DMA transfer is used to write 1 block of data to the SD buffer.

If the CPU is used to write data to the SDBUFR register, set the BWE flag to 0 before writing the data size <sup>\*1</sup> set in the SDSIZE.LEN[9:0] bits.

Note 1. If the transfer data size set in the SDSIZE.LEN[9:0] bits is an odd number, the odd numbered byte is ignored. Refer to section 48.5.2, SDBUFR Register Illegal Write Error for details.

**SDCLKCREN Flag (SDCLKCR Write Enable Flag)**

When a value is written to the SDCMD register, the SDHI starts the command sequence, the SDSTS2.CBSY flag becomes 1, and the SDSTS2.SDCLKCREN flag becomes 0. When the command sequence is complete, after the SDSTS2.CBSY flag becomes 0, eight cycles of the SDHI clock elapse and then the SDSTS2.SDCLKCREN flag becomes 1.

**ILA Flag (Illegal Access Error Detection Flag)**

— This flag becomes 1 under any of the following conditions:

- A value is written to the SDCMD register when the SDSTS2.CBSY flag is 1.
- The SDCMD.CMDTP bit is set to 1 (command accompanying data transfer), the SDCMD.ACMD[1:0] bits are set to 00b, and the SDCMD.CMDIDX[5:0] bits are set to 001100b (CMD12).

— This flag becomes 0 under the following condition:

- The flag is set to 0.

## 48.2.8 SD Interrupt Mask Register 1 (SDIMSK1)

Address(es): SDHI.SDIMSK1 0008 AC40h

|                    |     |     |     |         |         |        |         |         |
|--------------------|-----|-----|-----|---------|---------|--------|---------|---------|
|                    | b31 | b30 | b29 | b28     | b27     | b26    | b25     | b24     |
|                    | —   | —   | —   | —       | —       | —      | —       | —       |
| Value after reset: | 0   | 0   | 0   | 0       | 0       | 0      | 0       | 0       |
|                    | b23 | b22 | b21 | b20     | b19     | b18    | b17     | b16     |
|                    | —   | —   | —   | —       | —       | —      | —       | —       |
| Value after reset: | 0   | 0   | 0   | 0       | 0       | 0      | 0       | 0       |
|                    | b15 | b14 | b13 | b12     | b11     | b10    | b9      | b8      |
|                    | —   | —   | —   | —       | —       | —      | SDD3INM | SDD3RMM |
| Value after reset: | 0   | 0   | 0   | 0       | 0       | 0      | 1       | 1       |
|                    | b7  | b6  | b5  | b4      | b3      | b2     | b1      | b0      |
|                    | —   | —   | —   | SDCDINM | SDCDRMM | ACENDM | —       | RSPENDM |
| Value after reset: | 0   | 0   | 0   | 1       | 1       | 1      | 0       | 1       |

| Bit        | Symbol  | Bit Name                                 | Description                                                                                                                              | R/W |
|------------|---------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0         | RSPENDM | Response End Interrupt Request Mask      | 0: Response end interrupt request is not masked<br>1: Response end interrupt request is masked                                           | R/W |
| b1         | —       | Reserved                                 | This bit is 0 when read and cannot be modified.                                                                                          | R   |
| b2         | ACENDM  | Access End Interrupt Request Mask        | 0: Access end interrupt request is not masked<br>1: Access end interrupt request is masked                                               | R/W |
| b3         | SDCDRMM | SDHI_CD Removal Interrupt Request Mask   | 0: SD card removal interrupt request by the SDHI_CD pin not masked<br>1: SD card removal interrupt request by the SDHI_CD pin masked     | R/W |
| b4         | SDCDINM | SDHI_CD Insertion Interrupt Request Mask | 0: SD card insertion interrupt request by the SDHI_CD pin not masked<br>1: SD card insertion interrupt request by the SDHI_CD pin masked | R/W |
| b7 to b5   | —       | Reserved                                 | These bits are 0 when read and cannot be modified.                                                                                       | R   |
| b8         | SDD3RMM | SDHI_D3 Removal Interrupt Request Mask   | 0: SD card removal interrupt request by the SDHI_D3 pin not masked<br>1: SD card removal interrupt request by the SDHI_D3 pin masked     | R/W |
| b9         | SDD3INM | SDHI_D3 Insertion Interrupt Request Mask | 0: SD card insertion interrupt request by the SDHI_D3 pin not masked<br>1: SD card insertion interrupt request by the SDHI_D3 pin masked | R/W |
| b31 to b10 | —       | Reserved                                 | These bits are 0 when read and cannot be modified.                                                                                       | R   |

The SDIMSK1 register enables and disables the interrupt requests from the status flags in the SDSTS1 register. Refer to Table 48.8, Interrupt Sources for details on the relationship between the status flags and the requested interrupt source.

## 48.2.9 SD Interrupt Mask Register 2 (SDIMSK2)

Address(es): SDHI.SDIMSK2 0008 AC44h

|                    |        |      |      |       |       |       |       |
|--------------------|--------|------|------|-------|-------|-------|-------|
| b31                | b30    | b29  | b28  | b27   | b26   | b25   | b24   |
| —                  | —      | —    | —    | —     | —     | —     | —     |
| Value after reset: | 0      | 0    | 0    | 0     | 0     | 0     | 0     |
| b23                | b22    | b21  | b20  | b19   | b18   | b17   | b16   |
| —                  | —      | —    | —    | —     | —     | —     | —     |
| Value after reset: | 0      | 0    | 0    | 0     | 0     | 0     | 0     |
| b15                | b14    | b13  | b12  | b11   | b10   | b9    | b8    |
| ILAM               | —      | —    | —    | —     | —     | BWEM  | BREM  |
| Value after reset: | 1      | 0    | 0    | 0     | 1     | 0     | 1     |
| b7                 | b6     | b5   | b4   | b3    | b2    | b1    | b0    |
| —                  | RSPTOM | ILRM | ILWM | DTTOM | ENDEM | CRCEM | CMDEM |
| Value after reset: | 0      | 1    | 1    | 1     | 1     | 1     | 1     |

| Bit        | Symbol | Bit Name                                             | Description                                                                                                                                                    | R/W |
|------------|--------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0         | CMDEM  | Command Error Interrupt Request Mask                 | 0: Command error interrupt request not masked<br>1: Command error interrupt request masked                                                                     | R/W |
| b1         | CRCEM  | CRC Error Interrupt Request Mask                     | 0: CRC error interrupt request not masked<br>1: CRC error interrupt request masked                                                                             | R/W |
| b2         | ENDEM  | End Bit Error Interrupt Request Mask                 | 0: End bit detection error interrupt request not masked<br>1: End bit detection error interrupt request masked                                                 | R/W |
| b3         | DTTOM  | Data Timeout Interrupt Request Mask                  | 0: Data timeout interrupt request not masked<br>1: Data timeout interrupt request masked                                                                       | R/W |
| b4         | ILWM   | SDBUFR Register Illegal Write Interrupt Request Mask | 0: Illegal write detection interrupt request for the SDBUFR register not masked<br>1: Illegal write detection interrupt request for the SDBUFR register masked | R/W |
| b5         | ILRM   | SDBUFR Register Illegal Read Interrupt Request Mask  | 0: Illegal read detection interrupt request for the SDBUFR register not masked<br>1: Illegal read detection interrupt request for the SDBUFR register masked   | R/W |
| b6         | RSPTOM | Response Timeout Interrupt Request Mask              | 0: Response timeout interrupt request not masked<br>1: Response timeout interrupt request masked                                                               | R/W |
| b7         | —      | Reserved                                             | This bit is 0 when read and cannot be modified.                                                                                                                | R   |
| b8         | BREM   | BRE Interrupt Request Mask                           | 0: Read enable interrupt request for the SDBUFR register not masked<br>1: Read enable interrupt request for the SDBUFR register masked                         | R/W |
| b9         | BWEM   | BWE Interrupt Request Mask                           | 0: Write enable interrupt request for the SDBUFR register not masked<br>1: Write enable interrupt request for the SDBUFR register masked                       | R/W |
| b10        | —      | Reserved                                             | This bit is 0 when read and cannot be modified.                                                                                                                | R   |
| b11        | —      | Reserved                                             | This bit is 1 when read and cannot be modified.                                                                                                                | R   |
| b14 to b12 | —      | Reserved                                             | These bits are 0 when read and cannot be modified.                                                                                                             | R   |
| b15        | ILAM   | Illegal Access Error Interrupt Request Mask          | 0: Illegal access error interrupt request not masked<br>1: Illegal access error interrupt request masked                                                       | R/W |
| b31 to b16 | —      | Reserved                                             | These bits are 0 when read and cannot be modified.                                                                                                             | R   |

Note 1. When the SDIMSK2.BWEM bit is 0 or the SDIMSK2.BREM bit is 0, set the SDDMAEN.DMAEN bit to 0. When the SDDMAEN.DMAEN bit is 1, set the SDIMSK2.BWEM bit to 1 and the SDIMSK2.BREM bit to 1.

The SDIMSK2 register enables and disables the interrupt requests from the status flags in the SDSTS2 register. Refer to Table 48.8 for details on the relationship between the status flags and the requested interrupt source.

## 48.2.10 SDHI Clock Control Register (SDCLKCR)

Address(es): SDHI.SDCLKCR 0008 AC48h

|                    |             |     |     |     |     |     |            |       |
|--------------------|-------------|-----|-----|-----|-----|-----|------------|-------|
|                    | b31         | b30 | b29 | b28 | b27 | b26 | b25        | b24   |
|                    | —           | —   | —   | —   | —   | —   | —          | —     |
| Value after reset: | 0           | 0   | 0   | 0   | 0   | 0   | 0          | 0     |
|                    | b23         | b22 | b21 | b20 | b19 | b18 | b17        | b16   |
|                    | —           | —   | —   | —   | —   | —   | —          | —     |
| Value after reset: | 0           | 0   | 0   | 0   | 0   | 0   | 0          | 0     |
|                    | b15         | b14 | b13 | b12 | b11 | b10 | b9         | b8    |
|                    | —           | —   | —   | —   | —   | —   | CLKCTRLLEN | CLKEN |
| Value after reset: | 0           | 0   | 0   | 0   | 0   | 0   | 0          | 0     |
|                    | b7          | b6  | b5  | b4  | b3  | b2  | b1         | b0    |
|                    | CLKSEL[7:0] |     |     |     |     |     |            |       |
| Value after reset: | 0           | 0   | 1   | 0   | 0   | 0   | 0          | 0     |

| Bit        | Symbol      | Bit Name                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                 | R/W |
|------------|-------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b7 to b0   | CLKSEL[7:0] | SDHI Clock Frequency Select *1             | b7 b0<br>0 0 0 0 0 0 0: PCLKB divided by 2<br>0 0 0 0 0 0 1: PCLKB divided by 4<br>0 0 0 0 0 1 0: PCLKB divided by 8<br>0 0 0 0 1 0 0: PCLKB divided by 16<br>0 0 0 1 0 0 0: PCLKB divided by 32<br>0 0 1 0 0 0 0: PCLKB divided by 64<br>0 0 1 0 0 0 0: PCLKB divided by 128<br>0 1 0 0 0 0 0: PCLKB divided by 256<br>1 0 0 0 0 0 0: PCLKB divided by 512<br>1 1 1 1 1 1 1: PCLKB *2<br>Only set the values listed above. | R/W |
| b8         | CLKEN       | SDHI Clock Output Control *1               | 0: SDHI clock output is disabled (SDHI_CLK signal fixed low)<br>1: SDHI clock output enabled                                                                                                                                                                                                                                                                                                                                | R/W |
| b9         | CLKCTRLLEN  | SDHI Clock Output Automatic Control Select | 0: Automatic control of SDHI clock output disabled<br>1: Automatic control of SDHI clock output enabled                                                                                                                                                                                                                                                                                                                     | R/W |
| b31 to b10 | —           | Reserved                                   | These bits are 0 when read and cannot be modified.                                                                                                                                                                                                                                                                                                                                                                          | R   |

Note 1. Bits CLKSEL[7:0] and CLKEN cannot be write accessed when the SDSTS2.SDCLKCREN flag is 0.

Note 2. When setting the CLKSEL[7:0] bits to 1111111b or when changing the CLKSEL[7:0] bit values from 1111111b to another value, perform the following steps:

- (1) Set the CLKEN bit to 0. Do not change the other bit values.
- (2) Change the CLKSEL[7:0] bit values. Do not change the other bit values.
- (3) Set the CLKEN bit to 1. Do not change the other bit values.

The SDCLKCR register controls the SDHI clock frequency settings and output. Set the CLKEN bit to 1 before writing to the SDCMD register to start a command sequence. Do not write access the SDCLKCR register when the SDSTS2.SDCLKCREN flag is 0.

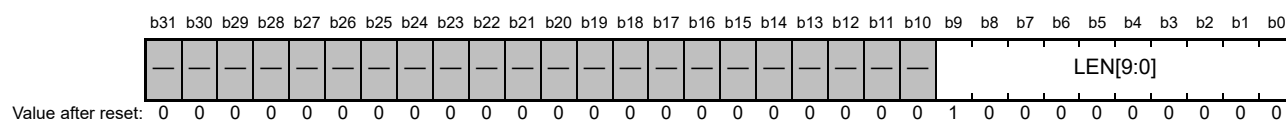
**CLKCTRLLEN Bit (SDHI Clock Output Automatic Control Select)**

The SDHI clock output automatic control is a function for starting and stopping SDHI clock output only during a command sequence. When this function is enabled, the SDHI starts outputting the SDHI clock after a value is set to the SDCMD register. After the command sequence is complete and eight cycles of the SDHI clock elapse, the SDHI stops outputting the SDHI clock. When the SDCLKCR.CLKEN bit is 0, output from the SDHI\_CLK pin becomes low regardless of the CLKCTRLLEN bit setting.



### 48.2.11 Transfer Data Size Register (SDSIZE)

Address(es): SDHI.SDSIZE 0008 AC4Ch



| Bit        | Symbol   | Bit Name                   | Description                                             | R/W |
|------------|----------|----------------------------|---------------------------------------------------------|-----|
| b9 to b0   | LEN[9:0] | Transfer Data Size Setting | Set the transfer data size. *1                          | R/W |
| b11 to b10 | —        | Reserved                   | These bits are 0 when read. Set them to 0 when writing. | R   |
| b31 to b12 | —        | Reserved                   | These bits are 0 when read and cannot be modified.      | R   |

Note 1. Do not rewrite these bits when the SDSTS2.CBSY flag is 1.

The SDSIZE register is used to set the transfer data size.

#### LEN[9:0] Bits (Transfer Data Size Setting)

When using single block transfer, the transfer data size can be set from 1 byte to 512 bytes. When CMD12 is automatically issued during a multi-block transfer sequence (CMD18 and CMD25), the transfer data size can only be set to 512 bytes. When CMD12 is not automatically issued during a multi-block transfer sequence, the transfer data size can be set to 32, 64, 128, 256, or 512 bytes. However, a 32-, 64-, 128-, or 256-byte multi-block read transfer can only be performed during an SDIO multi-block transfer (CMD53). Do not set these bits to 0 when using a command that includes data transfer.

## 48.2.12 Card Access Option Register (SDOPT)

Address SDHI.SDOPT 0008 AC50h

|                   |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| b31               | b30 | b29 | b28 | b27 | b26 | b25 | b24 | b23 | b22 | b21 | b20 | b19 | b18 | b17 | b16 |
| —                 | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   |
| Value after reset | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   |

|                   |     |     |     |     |     |    |    |          |    |    |    |           |    |    |    |
|-------------------|-----|-----|-----|-----|-----|----|----|----------|----|----|----|-----------|----|----|----|
| b15               | b14 | b13 | b12 | b11 | b10 | b9 | b8 | b7       | b6 | b5 | b4 | b3        | b2 | b1 | b0 |
| WIDTH             | —   | —   | —   | —   | —   | —  | —  | TOP[3:0] |    |    |    | CTOP[3:0] |    |    |    |
| Value after reset | 0   | 1   | 0   | 0   | 0   | 0  | 0  | 1        | 1  | 1  | 0  | 1         | 1  | 1  | 0  |

| Bit        | Symbol    | Bit Name                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | R/W |
|------------|-----------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b3 to b0   | CTOP[3:0] | Card Detection Time Counter *1 | <div> <div>b3 b0</div> <div>0 0 0 0: PCLKB × 2<sup>10</sup></div> <div>0 0 0 1: PCLKB × 2<sup>11</sup></div> <div>0 0 1 0: PCLKB × 2<sup>12</sup></div> <div>0 0 1 1: PCLKB × 2<sup>13</sup></div> <div>0 1 0 0: PCLKB × 2<sup>14</sup></div> <div>0 1 0 1: PCLKB × 2<sup>15</sup></div> <div>0 1 1 0: PCLKB × 2<sup>16</sup></div> <div>0 1 1 1: PCLKB × 2<sup>17</sup></div> </div> <div> <div>b3 b0</div> <div>1 0 0 0: PCLKB × 2<sup>18</sup></div> <div>1 0 0 1: PCLKB × 2<sup>19</sup></div> <div>1 0 1 0: PCLKB × 2<sup>20</sup></div> <div>1 0 1 1: PCLKB × 2<sup>21</sup></div> <div>1 1 0 0: PCLKB × 2<sup>22</sup></div> <div>1 1 0 1: PCLKB × 2<sup>23</sup></div> <div>1 1 1 0: PCLKB × 2<sup>24</sup></div> <div>1 1 1 1: Do not set this value.</div> </div>                                                                            | R/W |
| b7 to b4   | TOP[3:0]  | Timeout Counter *1             | <div> <div>b7 b4</div> <div>0 0 0 0: SDHI clock × 2<sup>13</sup></div> <div>0 0 0 1: SDHI clock × 2<sup>14</sup></div> <div>0 0 1 0: SDHI clock × 2<sup>15</sup></div> <div>0 0 1 1: SDHI clock × 2<sup>16</sup></div> <div>0 1 0 0: SDHI clock × 2<sup>17</sup></div> <div>0 1 0 1: SDHI clock × 2<sup>18</sup></div> <div>0 1 1 0: SDHI clock × 2<sup>19</sup></div> <div>0 1 1 1: SDHI clock × 2<sup>20</sup></div> </div> <div> <div>b7 b4</div> <div>1 0 0 0: SDHI clock × 2<sup>21</sup></div> <div>1 0 0 1: SDHI clock × 2<sup>22</sup></div> <div>1 0 1 0: SDHI clock × 2<sup>23</sup></div> <div>1 0 1 1: SDHI clock × 2<sup>24</sup></div> <div>1 1 0 0: SDHI clock × 2<sup>25</sup></div> <div>1 1 0 1: SDHI clock × 2<sup>26</sup></div> <div>1 1 1 0: SDHI clock × 2<sup>27</sup></div> <div>1 1 1 1: Do not set this value.</div> </div> | R/W |
| b8         | —         | Reserved                       | This bit is 0 when read. Set it to 0 when writing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R/W |
| b12 to b9  | —         | Reserved                       | These bits are 0 when read and cannot be modified.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R   |
| b13        | —         | Reserved                       | This bit is 0 when read. Set it to 0 when writing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R/W |
| b14        | —         | Reserved                       | This bit is 1 when read and cannot be modified.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R   |
| b15        | WIDTH     | SD Bus Width Select *1         | 0: Wide bus mode (4 bits)<br>1: Default bus mode (1 bit)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | R/W |
| b31 to b16 | —         | Reserved                       | These bits are 0 when read and cannot be modified.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | R   |

Note 1. Do not rewrite these bits when the SDSTS2.CBSY flag is 1.

The SD bus width and timeout counter are set in the SDOPT register.

## 48.2.13 SD Error Status Register 1 (SDERSTS1)

Address(es): SDHI.SDERSTS1 0008 AC58h

|                    |     |            |        |          |          |          |          |
|--------------------|-----|------------|--------|----------|----------|----------|----------|
| b31                | b30 | b29        | b28    | b27      | b26      | b25      | b24      |
| —                  | —   | —          | —      | —        | —        | —        | —        |
| Value after reset: | 0   | 0          | 0      | 0        | 0        | 0        | 0        |
| b23                | b22 | b21        | b20    | b19      | b18      | b17      | b16      |
| —                  | —   | —          | —      | —        | —        | —        | —        |
| Value after reset: | 0   | 0          | 0      | 0        | 0        | 0        | 0        |
| b15                | b14 | b13        | b12    | b11      | b10      | b9       | b8       |
| —                  |     | CRCTK[2:0] |        | CRCTKE   | RDCRCE   | RSPCRCE1 | RSPCRCE0 |
| Value after reset: | 0   | 0          | 1      | 0        | 0        | 0        | 0        |
| b7                 | b6  | b5         | b4     | b3       | b2       | b1       | b0       |
| —                  | —   | CRCLENE    | RDLENE | RSPLENE1 | RSPLENE0 | CMDE1    | CMDE0    |
| Value after reset: | 0   | 0          | 0      | 0        | 0        | 0        | 0        |

| Bit        | Symbol     | Bit Name                           | Description                                                                                                                                                                                                                                               | R/W |
|------------|------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0         | CMDE0      | Command Error Flag 0               | 0: Command index field value for a command *1 response is error free<br>1: Command index field value for a command *1 response is in error                                                                                                                | R   |
| b1         | CMDE1      | Command Error Flag 1               | 0: Command index field value for a command *2 response is error free<br>1: Command index field value for a command *2 response is in error (by setting the SDCMD.CMDIDX[5:0] bits, the error that occurs by issuing CMD12 is indicated by the CMDE0 flag) | R   |
| b2         | RSPLENE0   | Response Length Error Flag 0       | 0: Command *1 response length is error free<br>1: Command *1 response length is in error                                                                                                                                                                  | R   |
| b3         | RSPLENE1   | Response Length Error Flag 1       | 0: Command *2 response length is error free<br>1: Command *2 response length is in error (by setting the SDCMD.CMDIDX[5:0] bits, the error that occurs by issuing CMD12 is indicated by the RSPLENE0 flag)                                                | R   |
| b4         | RDLENE     | Read Data Length Error Flag        | 0: Read data length error did not occur<br>1: Read data length error occurred                                                                                                                                                                             | R   |
| b5         | CRCLENE    | CRC Status Token Length Error Flag | 0: CRC status token length error did not occur<br>1: CRC status token length error occurred                                                                                                                                                               | R   |
| b7, b6     | —          | Reserved                           | These bits are 0 when read.                                                                                                                                                                                                                               | R   |
| b8         | RSPCRCE0   | Response CRC Error Flag 0          | 0: No CRC error detected in command *1 response<br>1: CRC error detected in command *1 response                                                                                                                                                           | R   |
| b9         | RSPCRCE1   | Response CRC Error Flag 1          | 0: No CRC error detected in command *2 response<br>1: CRC error detected in command *2 response (the error that occurs for CMD12 with the setting of the SDCMD.CMDIDX[5:0] bits is indicated by the RSPCRCE0 flag)                                        | R   |
| b10        | RDCRCE     | Read Data CRC Error Flag           | 0: No CRC error detected in read data<br>1: CRC error detected in read data                                                                                                                                                                               | R   |
| b11        | CRCTKE     | CRC Status Token Error Flag        | 0: No error detected in CRC status token<br>1: Error detected in CRC status token                                                                                                                                                                         | R   |
| b14 to b12 | CRCTK[2:0] | CRC Status Token                   | Store the CRC status token value (normal value is 010b)                                                                                                                                                                                                   | R   |
| b15        | —          | Reserved                           | This bit is 0 when read.                                                                                                                                                                                                                                  | R   |
| b31 to b16 | —          | Reserved                           | These bits are undefined when read.                                                                                                                                                                                                                       | R   |

Note 1. Command other than CMD12 or CMD52 which are automatically issued to stop data transfer.

Note 2. CMD12 or CMD52 which are automatically issued to stop data transfer.

The SDERSTS1 register indicates the CRC status token, CRC error, end bit error, and command error.

## 48.2.14 SD Error Status Register 2 (SDERSTS2)

Address(es): SDHI.SDERSTS2 0008 AC5Ch

|                    |     |          |       |      |        |        |        |        |
|--------------------|-----|----------|-------|------|--------|--------|--------|--------|
|                    | b31 | b30      | b29   | b28  | b27    | b26    | b25    | b24    |
|                    | —   | —        | —     | —    | —      | —      | —      | —      |
| Value after reset: | 0   | 0        | 0     | 0    | 0      | 0      | 0      | 0      |
|                    | b23 | b22      | b21   | b20  | b19    | b18    | b17    | b16    |
|                    | —   | —        | —     | —    | —      | —      | —      | —      |
| Value after reset: | 0   | 0        | 0     | 0    | 0      | 0      | 0      | 0      |
|                    | b15 | b14      | b13   | b12  | b11    | b10    | b9     | b8     |
|                    | —   | —        | —     | —    | —      | —      | —      | —      |
| Value after reset: | 0   | 0        | 0     | 0    | 0      | 0      | 0      | 0      |
|                    | b7  | b6       | b5    | b4   | b3     | b2     | b1     | b0     |
|                    | —   | CRCBSYTO | CRCTO | RDTO | BSYTO1 | BSYTO0 | RSPTO1 | RSPTO0 |
| Value after reset: | 0   | 0        | 0     | 0    | 0      | 0      | 0      | 0      |

| Bit       | Symbol   | Bit Name                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                   | R/W |
|-----------|----------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0        | RSPTO0   | Response Timeout Flag 0            | 0: After a command *1 was issued, a response was received in less than 640 cycles of the SDHI clock.<br>1: After a command *1 was issued, a response was not received even after 640 cycles or more of the SDHI clock elapsed.                                                                                                                                                                                                                | R   |
| b1        | RSPTO1   | Response Timeout Flag 1            | 0: After a command *2 was issued, a response was received in less than 640 cycles of the SDHI clock.<br>1: After a command *2 was issued, a response was not received even after 640 cycles or more of the SDHI clock elapsed (by setting the SDCMD.CMDIDX[5:0] bits, the error that occurs by issuing CMD12 is indicated by the RSPTO0 flag).                                                                                                | R   |
| b2        | BSYTO0   | Busy Timeout Flag 0                | 0: After the R1b response was received, the SDHI was released from the busy state during the specified period *3.<br>1: After the R1b response was received, the SDHI was in the busy state even after the specified period *3 elapsed.                                                                                                                                                                                                       | R   |
| b3        | BSYTO1   | Busy Timeout Flag 1                | 0: After CMD12 was automatically issued, the SDHI was released from the busy state during the specified period *3.<br>1: After CMD12 was automatically issued, the SDHI was in the busy state even after the specified period *3 elapsed (by setting the SDCMD.CMDIDX[5:0] bits, the error that occurs by issuing CMD12 is indicated by the BSYTO0 flag).                                                                                     | R   |
| b4        | RDTO     | Read Data Timeout Flag             | After a read command is issued, this flag becomes 1 when read data is not received even after the specified period *3 elapses.<br>After read data is received, this flag becomes 1 when the next block of read data is not received even after the specified period *3 elapses.<br>After the SDHI exits the read wait state, this flag becomes 1 when the next block of read data is not received even after the specified period *3 elapses. | R   |
| b5        | CRCTO    | CRC Status Token Timeout Flag      | 0: After data was written to the SD card, a CRC status token was received during the specified period *3.<br>1: After CRC data was written to the SD card, a CRC status token was not received even after the specified period *3 elapsed.                                                                                                                                                                                                    | R   |
| b6        | CRCBSYTO | CRC Status Token Busy Timeout Flag | 0: After a CRC status token was received, the SDHI was released from the busy state during the specified period *3.<br>1: After a CRC status token was received, the SDHI is in the busy state even after the specified period *3 elapsed.                                                                                                                                                                                                    | R   |
| b31 to b7 | —        | Reserved                           | These bits are 0 when read.                                                                                                                                                                                                                                                                                                                                                                                                                   | R   |

Note 1. Command other than CMD12 or CMD52 which are automatically issued to stop data transfer.

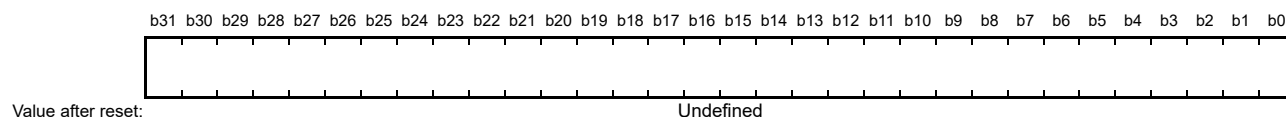
Note 2. CMD12 or CMD52 which are automatically issued to stop transfer.

Note 3. Set the SDOPT.TOP[3:0] bits to select the number of  $n$  cycles.

The SDERSTS2 register indicates the timeout status.

### 48.2.15 SD Buffer Register (SDBUFR)

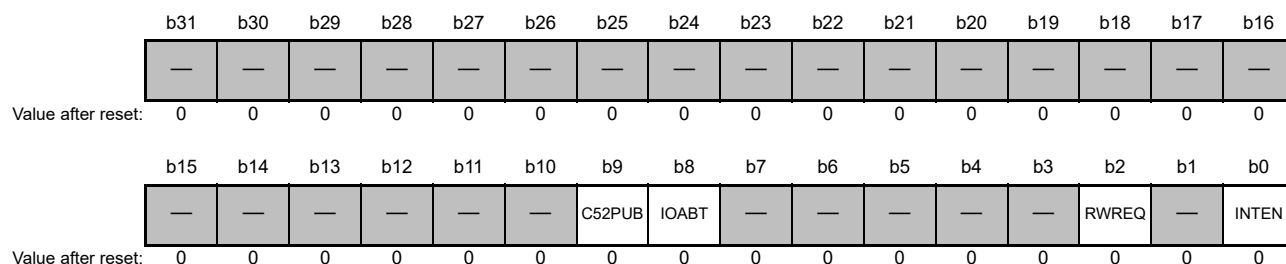
Address(es): SDHI.SDBUFR 0008 AC60h



The SDBUFR register is used when writing data to the SD card and when reading data from the SD card. The SDBUFR register is connected to the SDHI's internal SD buffer. Refer to section 48.3.1, Data Block Format of the SD Card for details on the configuration of the SDBUFR register and the SD buffer.

### 48.2.16 SDIO Mode Control Register (SDIOMD)

Address(es): SDHI.SDIOMD 0008 AC68h



| Bit        | Symbol | Bit Name                                       | Description                                                                                                                                                             | R/W |
|------------|--------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| b0         | INTEN  | SDIO Interrupt Acceptance Enable <sup>*1</sup> | 0: SDIO interrupt accept disabled<br>1: SDIO interrupt accept enabled                                                                                                   | R/W |
| b1         | —      | Reserved                                       | This bit is 0 when read and cannot be modified.                                                                                                                         | R   |
| b2         | RWREQ  | Read Wait Request                              | 0: SDHI exits read wait state<br>1: Request for SDHI to enter read wait state                                                                                           | R/W |
| b7 to b3   | —      | Reserved                                       | These bits are 0 when read and cannot be modified.                                                                                                                      | R   |
| b8         | IOABT  | SDIO Abort                                     | If this bit is set to 1 during multi-block transfer triggered by CMD53, CMD52 is immediately issued, and the command sequence is aborted.                               | R/W |
| b9         | C52PUB | SDIO None Abort                                | If this bit is set to 1 during multi-block transfer triggered by CMD53, CMD52 is issued before the transfer process is complete, and the command sequence is completed. | R/W |
| b31 to b10 | —      | Reserved                                       | These bits are 0 when read and cannot be modified.                                                                                                                      | R   |

Note 1. Do not rewrite this bit when the SDSTS2.CBSY flag is 1.

The SDIOMD register controls reception of the SDIO interrupt, controls CMD52 issuance during multi-block transfer, and controls the read wait request. Do not set bits C52PUB and IOABT to 1 at the same time.

**RWREQ Bit (Read Wait Request)**

If the RWREQ bit is set to 1 during a multi-block read sequence triggered by issuing CMD53, when the current block is done being read, the SDHI enters the read wait state. The method for exiting the read wait state is as follows.

- If the RWREQ bit is set to 0 while the SDHI is in the read wait state, the SDHI exits the read wait state.
- If the IOABT bit is set to 1 while the SDHI is in the read wait state, after CMD52 is issued, the RWREQ bit becomes 0 and the SDHI exits the read wait state.
- If bits C52PUB and RWREQ are simultaneously set to 1 during a multi-block read sequence triggered by issuing CMD53 \*1, the SDHI does not automatically exit the read wait state, so after receiving the CMD52 response, set the RWREQ bit to 0.

Note 1. Set bits RWREQ and C52PUB to 1 simultaneously.

If the RWREQ bit is set to 1 while the last block is being transferred during a multi-block read sequence triggered by issuing CMD53, the SDHI will not enter the read wait state, the SDSTS1.ACEND flag becomes 1, and the RWREQ bit becomes 0. Set the RWREQ bit to 1 after the SDSTS1.RSPEND flag becomes 1.

**IOABT Bit (SDIO Abort)**

- If the IOABT bit is set to 1 during a multi-block transfer sequence triggered by issuing CMD53, the SDHI stops the CMD53 command sequence, and CMD52 is issued. If the command sequence is stopped due to a communication error or timeout, the SDHI does not issue CMD52. The SD buffer can be accessed even after the IOABT bit is set to 1, but the SDSTS2.ILR flag or ILW flag becomes 1, and a buffer access error occurs. Write a value to the SDARG register before setting the IOABT bit to 1.
- During a single block write, if there is no data in the SD buffer when the IOABT bit is set to 1, the SDHI does not issue CMD52, and the SDSTS1.ACEND flag becomes 1. If there is data in the SD buffer when the IOABT bit is set to 1, the SDHI does not issue CMD52, and after the SDHI exits the busy state, the SDSTS1.ACEND flag becomes 1.
- If the IOABT bit is set to 1 during a single block read, the SDHI does not issue CMD52, and the SDSTS1.ACEND flag immediately becomes 1.
- If the SDHI is in the busy state after the R1b response is received and the IOABT bit is set to 1, the SDHI does not issue CMD52, and after the SDHI exits the busy state, the SDSTS1.ACEND flag becomes 1.
- If the IOABT bit is set to 1 after the command sequence is completed, the SDHI does not issue CMD52, and the SDSTS1.ACEND flag does not become 1.
- Set the IOABT bit to 1 after the SDSTS1.RSPEND flag becomes 1.
- Set the IOABT bit to 0 after the SDSTS1.ACEND flag becomes 1.

**C52PUB Bit (SDIO None Abort)**

- If the C52PUB bit is set to 1 during a multi-block write sequence triggered by issuing CMD53, CMD52 is automatically issued when the SD buffer is empty and the current block write access is complete. The C52PUB bit becomes 0 after the response for CMD52 is received. If the C52PUB bit is 1 while the last block is being transferred, the SDHI does not issue CMD52, and after the SDSTS1.RSPEND flag becomes 1, the C52PUB bit is set to 0.
- If the C52PUB bit and RWREQ bit are set to 1 during a multi-block read sequence triggered by issuing CMD53, the SDHI enters the read wait state after the current block read access is complete, and the SDHI automatically issues CMD52. The C52PUB bit becomes 0 after the response for CMD52 is received. If the C52PUB bit is set to 1 while the last block is being transferred, the SDHI does not issue CMD52, and after the SDSTS1.RSPEND flag becomes 1, the C52PUB bit is set to 0.
- During a multi-block read sequence triggered by issuing CMD53, if the C52PUB bit is set to 1, also set the RWREQ bit to 1.
- Write a value to the SDARG register before setting the C52PUB bit to 1.
- Set the C52PUB bit to 1 after the SDSTS1.RSPEND flag becomes 1.

## 48.2.17 SDIO Status Register (SDIOSTS)

Address(es): SDHI.SDIOSTS 0008 AC6Ch

|                    |      |         |     |     |     |     |     |       |
|--------------------|------|---------|-----|-----|-----|-----|-----|-------|
|                    | b31  | b30     | b29 | b28 | b27 | b26 | b25 | b24   |
|                    | —    | —       | —   | —   | —   | —   | —   | —     |
| Value after reset: | 0    | 0       | 0   | 0   | 0   | 0   | 0   | 0     |
|                    | b23  | b22     | b21 | b20 | b19 | b18 | b17 | b16   |
|                    | —    | —       | —   | —   | —   | —   | —   | —     |
| Value after reset: | 0    | 0       | 0   | 0   | 0   | 0   | 0   | 0     |
|                    | b15  | b14     | b13 | b12 | b11 | b10 | b9  | b8    |
|                    | EXWT | EXPUB52 | —   | —   | —   | —   | —   | —     |
| Value after reset: | 0    | 0       | 0   | 0   | 0   | 0   | 0   | 0     |
|                    | b7   | b6      | b5  | b4  | b3  | b2  | b1  | b0    |
|                    | —    | —       | —   | —   | —   | —   | —   | IOIRQ |
| Value after reset: | 0    | 0       | 0   | 0   | 0   | 0   | 0   | 0     |

| Bit        | Symbol  | Bit Name                   | Description                                                     | R/W      |
|------------|---------|----------------------------|-----------------------------------------------------------------|----------|
| b0         | IOIRQ   | SDIO Interrupt Status Flag | 0: SDIO interrupt not accepted<br>1: SDIO interrupt accepted    | R/(W) *1 |
| b2, b1     | —       | Reserved                   | These bits are undefined when read. Set them to 1 when writing. | R/W      |
| b13 to b3  | —       | Reserved                   | These bits are 0 when read and cannot be modified.              | R        |
| b14        | EXPUB52 | EXPUB52 Status Flag        | Indicates the status of the EXPUB52                             | R/(W) *1 |
| b15        | EXWT    | EXWT Status Flag           | Indicates the status of the EXWT                                | R/(W) *1 |
| b31 to b16 | —       | Reserved                   | These bits are 0 when read and cannot be modified.              | R        |

Note 1. The flag value does not change even when set to 1. If 0 is written to this flag, it becomes 0.

The SDIOSTS register indicates the status of the SDIO card access. When clearing a flag, bits to be cleared should be set to 0; bits not being cleared should be set to 1.

**IOIRQ Flag (SDIO Interrupt Status Flag)**

— This flag becomes 1 under the following condition:

- The SDIO interrupt from the SDIO card is accepted while the SDIOMD.INTEN bit is 1.

— This flag becomes 0 under the following condition:

- The flag is set to 0. \*1

Note 1. Access the SDIO card, negate the SDIO interrupt from the SDIO card, and then set the IOIRQ flag to 0. If the SDIO interrupt from the SDIO card is not negated, the IOIRQ flag might become 1 again.

**EXPUB52 Flag (EXPUB52 Status Flag)**

— This flag becomes 1 under any of the following conditions:

- When multi-block transfer is triggered by CMD53 being issued, the SDIOMD.C52PUB bit is set to 1 while the last block is being transferred.
- When multi-block write is triggered by CMD53 being issued, the SDIOMD.C52PUB bit remains set to 1 while the last block is being transferred.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

**EXWT Flag (EXWT Status Flag)**

— This flag becomes 1 under the following condition:

- During a multi-block read sequence triggered by CMD53 being issued, the SDIOMD.RWREQ bit is set to 1 while the last block is being transferred.

— This flag becomes 0 under the following condition:

- The flag is set to 0.

**48.2.18 SDIO Interrupt Mask Register (SDIOIMSK)**

Address(es): SDHI.SDIOIMSK 0008 AC70h

|                    |       |          |     |     |     |     |     |        |
|--------------------|-------|----------|-----|-----|-----|-----|-----|--------|
|                    | b31   | b30      | b29 | b28 | b27 | b26 | b25 | b24    |
|                    | —     | —        | —   | —   | —   | —   | —   | —      |
| Value after reset: | 0     | 0        | 0   | 0   | 0   | 0   | 0   | 0      |
|                    | b23   | b22      | b21 | b20 | b19 | b18 | b17 | b16    |
|                    | —     | —        | —   | —   | —   | —   | —   | —      |
| Value after reset: | 0     | 0        | 0   | 0   | 0   | 0   | 0   | 0      |
|                    | b15   | b14      | b13 | b12 | b11 | b10 | b9  | b8     |
|                    | EXWTM | EXPUB52M | —   | —   | —   | —   | —   | —      |
| Value after reset: | 1     | 1        | 0   | 0   | 0   | 0   | 0   | 0      |
|                    | b7    | b6       | b5  | b4  | b3  | b2  | b1  | b0     |
|                    | —     | —        | —   | —   | —   | —   | —   | IOIRQM |
| Value after reset: | 0     | 0        | 0   | 0   | 0   | 1   | 1   | 1      |

| Bit        | Symbol   | Bit Name                               | Description                                                                    | R/W |
|------------|----------|----------------------------------------|--------------------------------------------------------------------------------|-----|
| b0         | IOIRQM   | IOIRQ Interrupt Mask Control           | 0: IOIRQ interrupt not masked<br>1: IOIRQ interrupt masked                     | R/W |
| b2, b1     | —        | Reserved                               | These bits are 1 when read. Set them to 1 when writing.                        | R/W |
| b13 to b3  | —        | Reserved                               | These bits are 0 when read and cannot be modified.                             | R   |
| b14        | EXPUB52M | EXPUB52 Interrupt Request Mask Control | 0: EXPUB52 interrupt request not masked<br>1: EXPUB52 interrupt request masked | R/W |
| b15        | EXWTM    | EXWT Interrupt Request Mask Control    | 0: EXWT interrupt request not masked<br>1: EXWT interrupt request masked       | R/W |
| b31 to b16 | —        | Reserved                               | These bits are 0 when read and cannot be modified.                             | R   |

The SDIOIMSK register enables and disables the interrupt requests from the status flags in the SDIOSTS register. Refer to Table 48.8, Interrupt Sources for details on the relationship between the status flags and the requested interrupt source.



## 48.2.19 DMA Transfer Enable Register (SDDMAEN)

Address(es): SDHI.SDDMAEN 0008 ADB0h

|                    |     |     |     |     |     |     |     |     |     |     |     |     |     |       |     |
|--------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|-----|
| b31                | b30 | b29 | b28 | b27 | b26 | b25 | b24 | b23 | b22 | b21 | b20 | b19 | b18 | b17   | b16 |
| —                  | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —     | —   |
| Value after reset: | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   |
| b15                | b14 | b13 | b12 | b11 | b10 | b9  | b8  | b7  | b6  | b5  | b4  | b3  | b2  | b1    | b0  |
| —                  | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | DMAEN | —   |
| Value after reset: | 0   | 0   | 0   | 1   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 1   | 0   | 0     | 0   |

| Bit        | Symbol | Bit Name                  | Description                                                                                                                       | R/W |
|------------|--------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----|
| b0         | —      | Reserved                  | This bit is 0 when read. Set it to 0 when writing.                                                                                | R/W |
| b1         | DMAEN  | DMA Transfer Enable *1 *2 | 0: Using DMAC and DTC to access the SDBUFR register is disabled<br>1: Using DMAC and DTC to access the SDBUFR register is enabled | R/W |
| b3, b2     | —      | Reserved                  | These bits are 0 when read. Set them to 0 when writing.                                                                           | R   |
| b4         | —      | Reserved                  | This bit is 1 when read. Set it to 1 when writing.                                                                                | R   |
| b5         | —      | Reserved                  | This bit is 0 when read. Set it to 0 when writing.                                                                                | R/W |
| b7, b6     | —      | Reserved                  | These bits are 0 when read and cannot be modified.                                                                                | R   |
| b9, b8     | —      | Reserved                  | These bits are 0 when read. Set them to 0 when writing.                                                                           | R/W |
| b11, b10   | —      | Reserved                  | These bits are 0 when read and cannot be modified.                                                                                | R   |
| b12        | —      | Reserved                  | This bit is 1 when read. Set it to 1 when writing.                                                                                | R   |
| b31 to b13 | —      | Reserved                  | These bits are 0 when read and cannot be modified.                                                                                | R   |

Note 1. Do not rewrite this bit when the SDSTS2.CBSY bit is 1.

Note 2. When the SDIMSK2.BWEM bit is 0 or the SDIMSK2.BREM bit is 0, set the SDDMAEN.DMAEN bit to 0. When the SDDMAEN.DMAEN bit is 1, set the SDIMSK2.BWEM bit to 1 and the SDIMSK2.BREM bit to 1.

The SDDMAEN register enables and disables DMA transfer.

**DMAEN Bit (DMA Transfer Enable)**

When using DMA transfer to access the SD buffer, set the DMAEN bit to 1 before setting the SDCMD register.

## 48.2.20 SDHI Software Reset Register (SDRST)

Address(es): SDHI.SDRST 0008 ADC0h

|                    |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|--------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|                    | b31 | b30 | b29 | b28 | b27 | b26 | b25 | b24 | b23 | b22 | b21 | b20 | b19 | b18 | b17 | b16 |
|                    | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   |
| Value after reset: | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   |

|                    |     |     |     |     |     |     |    |    |    |    |    |    |    |    |    |       |
|--------------------|-----|-----|-----|-----|-----|-----|----|----|----|----|----|----|----|----|----|-------|
|                    | b15 | b14 | b13 | b12 | b11 | b10 | b9 | b8 | b7 | b6 | b5 | b4 | b3 | b2 | b1 | b0    |
|                    | —   | —   | —   | —   | —   | —   | —  | —  | —  | —  | —  | —  | —  | —  | —  | SDRST |
| Value after reset: | 0   | 0   | 0   | 0   | 0   | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 1     |

| Bit       | Symbol | Bit Name                    | Description                                               | R/W |
|-----------|--------|-----------------------------|-----------------------------------------------------------|-----|
| b0        | SDRST  | SDHI Software Reset Control | 0: SDHI software reset<br>1: SDHI software reset released | R/W |
| b2, b1    | —      | Reserved                    | These bits are 1 when read. Set them to 1 when writing.   | R   |
| b31 to b3 | —      | Reserved                    | These bits are 0 when read and cannot be modified.        | R   |

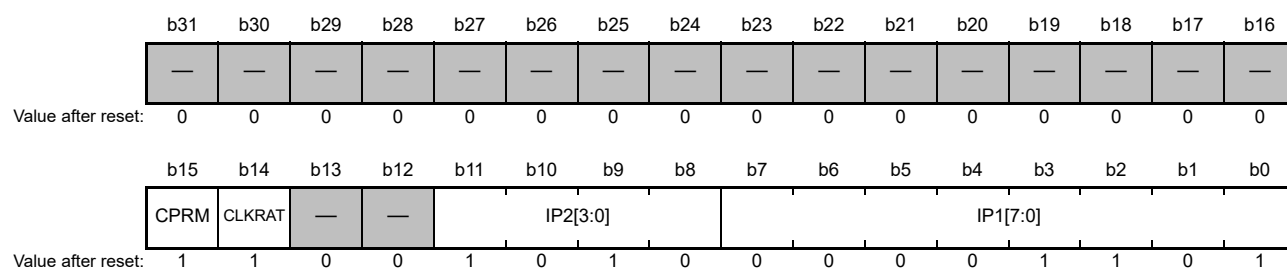
Table 48.5 lists the bits and flags initialized by the SDHI software reset.

**Table 48.5 Bits and Flags Initialized by the SDHI Software Reset**

| Register | Bit/Flag                                                                                                             |
|----------|----------------------------------------------------------------------------------------------------------------------|
| SDSTOP   | SDBLKCNTEN                                                                                                           |
| SDSTS1   | RSPEND, ACEND                                                                                                        |
| SDSTS2   | CMDE, CRCE, ENDE, DTO, ILW, ILR, RSPTO, SDD0MON, BRE, BWE, SDCLKCREN, ILA                                            |
| SDCLKCR  | CLKEN                                                                                                                |
| SDOPT    | CTOP[3:0], TOP[3:0], WIDTH<br>Bits b8 and b13 in the SDOPT register are also initialized by the SDHI software reset. |
| SDERSTS1 | CMDE0, CMDE1, RSPLNE0, RSPLNE1, RDLENE, CRCLNE, RSPCRCE0, RSPCRCE1, RDCRCE, CRCTKE, CRCTK[2:0]                       |
| SDERSTS2 | RSPTO0, RSPTO1, BSYTO0, BSYTO1, RDTO, CRCTO, CRCBSYTO                                                                |
| SDIOSTS  | IOIRQ, EXPUB52, EXWT                                                                                                 |

## 48.2.21 Version Register (SDVER)

Address(es): SDHI.SDVER 0008 ADC4h



| Bit        | Symbol   | Bit Name                  | Description                                                                                                    | R/W |
|------------|----------|---------------------------|----------------------------------------------------------------------------------------------------------------|-----|
| b7 to b0   | IP1[7:0] | IP Version 1              | IP version 1                                                                                                   | R   |
| b11 to b8  | IP2[3:0] | IP Version 2              | IP version 2                                                                                                   | R   |
| b13, b12   | —        | Reserved                  | These bits are 0 when read.                                                                                    | R   |
| b14        | CLKRAT   | Operating Clock Condition | 0: SDHI clock frequency = PCLKB frequency not supported<br>1: SDHI clock frequency = PCLKB frequency supported | R   |
| b15        | CPRM     | CPRM Function Select      | 0: CPRM function is enabled<br>1: CPRM function is disabled                                                    | R   |
| b31 to b16 | —        | Reserved                  | These bits are 0 when read.                                                                                    | R   |

The SDVER register indicates the SDHI version.

## 48.2.22 Swap Control Register (SDSWAP)

Address(es): SDHI.SDSWAP 0008 ADE0h

|                    |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|--------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|                    | b31 | b30 | b29 | b28 | b27 | b26 | b25 | b24 | b23 | b22 | b21 | b20 | b19 | b18 | b17 | b16 |
|                    | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   | —   |
| Value after reset: | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   |

|                    |     |     |     |     |     |     |    |    |       |       |    |    |    |    |    |    |
|--------------------|-----|-----|-----|-----|-----|-----|----|----|-------|-------|----|----|----|----|----|----|
|                    | b15 | b14 | b13 | b12 | b11 | b10 | b9 | b8 | b7    | b6    | b5 | b4 | b3 | b2 | b1 | b0 |
|                    | —   | —   | —   | —   | —   | —   | —  | —  | BRSWP | BWSWP | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0   | 0   | 0   | 0   | 0   | 0   | 0  | 0  | 0     | 0     | 0  | 0  | 0  | 0  | 0  | 0  |

| Bit        | Symbol | Bit Name             | Description                                                                                | R/W |
|------------|--------|----------------------|--------------------------------------------------------------------------------------------|-----|
| b0         | —      | Reserved             | This bit is 0 when read and cannot be modified.                                            | R   |
| b1         | —      | Reserved             | This bit is 0 when read. Set it to 0 when writing.                                         | R/W |
| b2         | —      | Reserved             | This bit is 0 when read and cannot be modified.                                            | R   |
| b4, b3     | —      | Reserved             | These bits are 0 when read. Set them to 0 when writing.                                    | R/W |
| b5         | —      | Reserved             | This bit is 0 when read and cannot be modified.                                            | R   |
| b6         | BWSWP  | SDBUFR Swap Write *1 | 0: Normal write operation<br>1: Swap the byte endian before writing to the SDBUFR register | R/W |
| b7         | BRSWP  | SDBUFR Swap Read *1  | 0: Normal read operation<br>1: Swap the byte endian before reading the SDBUFR register     | R/W |
| b10 to b8  | —      | Reserved             | These bits are 0 when read and cannot be modified.                                         | R   |
| b12, b11   | —      | Reserved             | These bits are 0 when read. Set them to 0 when writing.                                    | R/W |
| b14, b13   | —      | Reserved             | These bits are 0 when read and cannot be modified.                                         | R   |
| b15        | —      | Reserved             | This bit is 0 when read. Set it to 0 when writing.                                         | R/W |
| b31 to b16 | —      | Reserved             | These bits are 0 when read and cannot be modified.                                         | R   |

Note 1. Do not rewrite this bit when the SDSTS2.CBSY flag is 1.

The SDSWAP register is used to select whether or not the byte endian is swapped when accessing the SDBUFR register. Refer to section 48.3.1 for details on the differences in accessing the SDBUFR register based on the SDSWAP register value.

48.3 SDHI Operation

48.3.1 Data Block Format of the SD Card

The SDHI has a default bus mode (1-bit width) that uses just the SDHI\_D0 pin as a data line and a wide bus mode (4-bit width) that uses pins SDHI\_D0 to SDHI\_D3. Figure 48.2 shows the transfer format when the SDOPT.WIDTH bit is 1 (default bus mode), and Figure 48.3 shows the transfer format when the SDOPT.WIDTH bit is 0 (wide bus mode).

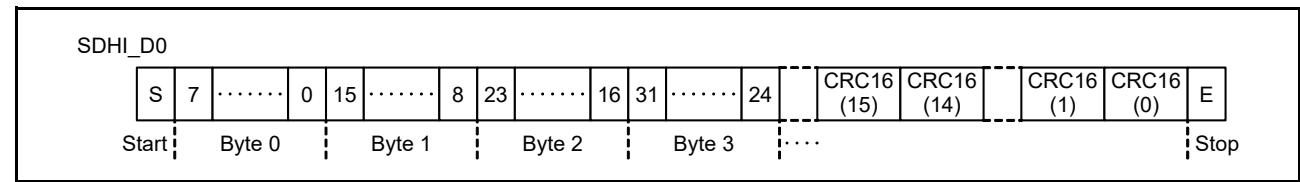


Figure 48.2 Transfer Format in Default Bus Mode

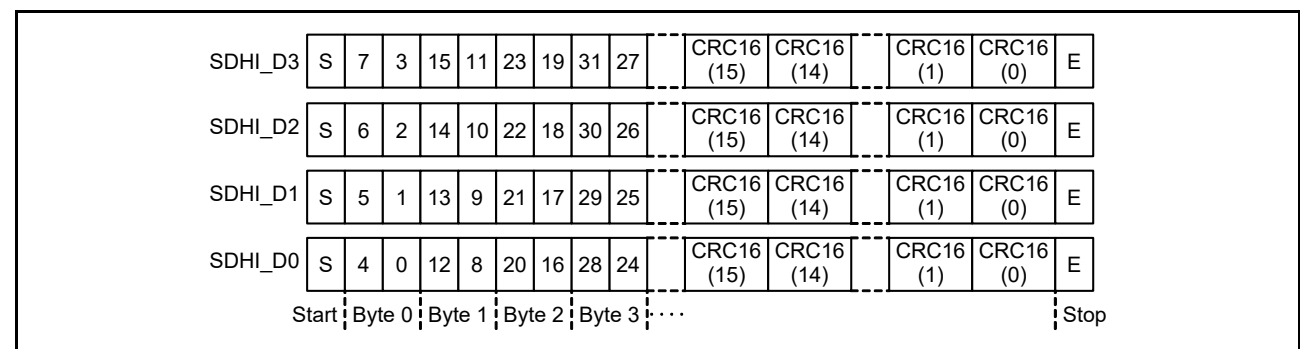
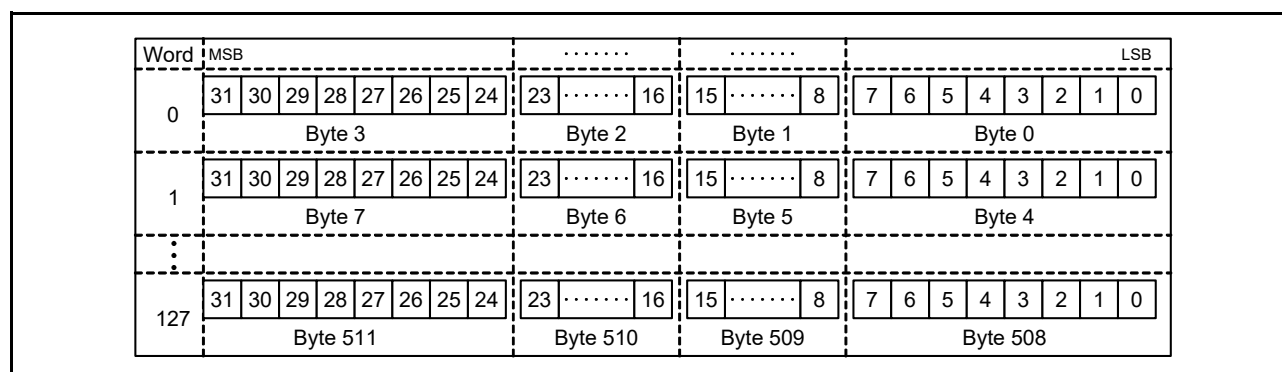


Figure 48.3 Transfer Format in Wide Bus Mode

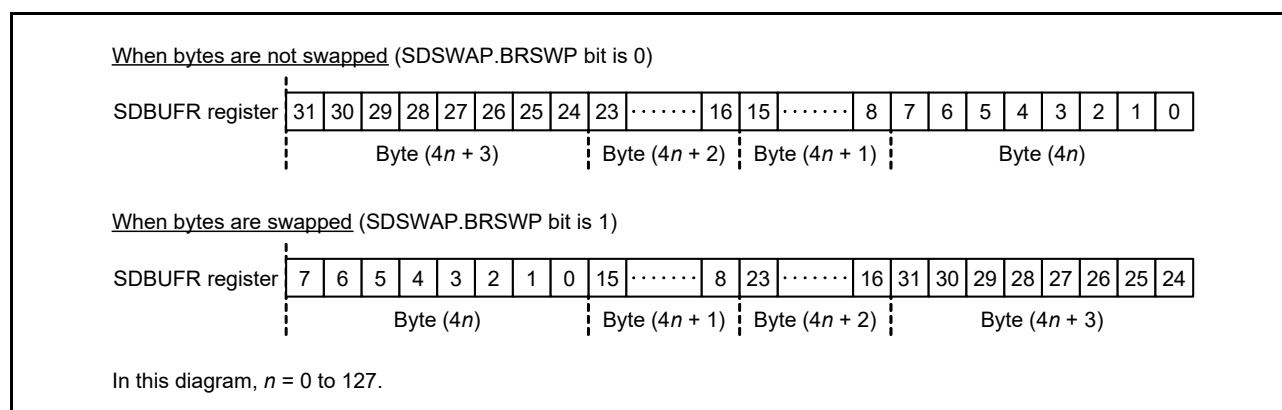
### 48.3.2 SD Buffer and the SDBUFR Register

The SDHI transfers data to an SD card via its internal SD buffer. The SD buffer is comprised of a double buffer, and each buffer is 512 bytes. Figure 48.4 shows the data configuration of a single buffer of the SD buffer's double buffer.



**Figure 48.4** Data Configuration of Single Buffer in the SD Buffer

Access to the SD buffer is done via the SDBUFR register. If data is written to the SDBUFR register while the SDSWAP.BSWP bit is 1, the SDHI swaps the endian for the byte, and stores the data in the SDBUFR register. If data is read from the SDBUFR register while the SDSWAP.BRSWP bit is 1, the data of the endian for the byte that was swapped can be read. Figure 48.5 shows the data alignment when reading the SDBUFR register.



**Figure 48.5** Data Alignment When Reading the SDBUFR Register

### 48.3.3 SD Card Detection

The SDHI can detect an SD card using either the SDHI\_CD pin or SDHI\_D3 pin.

#### 48.3.3.1 Using the SDHI\_CD Pin to Detect an SD Card

Figure 48.6 shows the timing chart for SD card detection using the SDHI\_CD pin. The SDHI\_CD pin is connected to the card detection switch of the SD card connector, and is pulled-up by the MCU. The pull-up resistance value is determined by the specifications of the host device. Note that there are some SD card sockets whose card detection switches become open when the SD card is inserted.

- Detecting SD card insertion

The signal from the SDHI\_CD pin becomes low when an SD card is inserted. This causes the SDSTS1.SDCDIN flag to become 1 if the SDHI\_CD pin is low for the number of cycles set in the SDOPT.CTOP[3:0] bits. The SDSTS1.SDCDIN flag is cleared by setting it to 0.

- Detecting SD card removal

The signal from the SDHI\_CD pin becomes high when the SD card is removed. This causes the SDSTS1.SDCDRM flag to become 1 if the SDHI\_CD pin is high for the number of cycles set in the SDOPT.CTOP[3:0] bits. The SDSTS1.SDCDRM flag is cleared by setting it to 0.

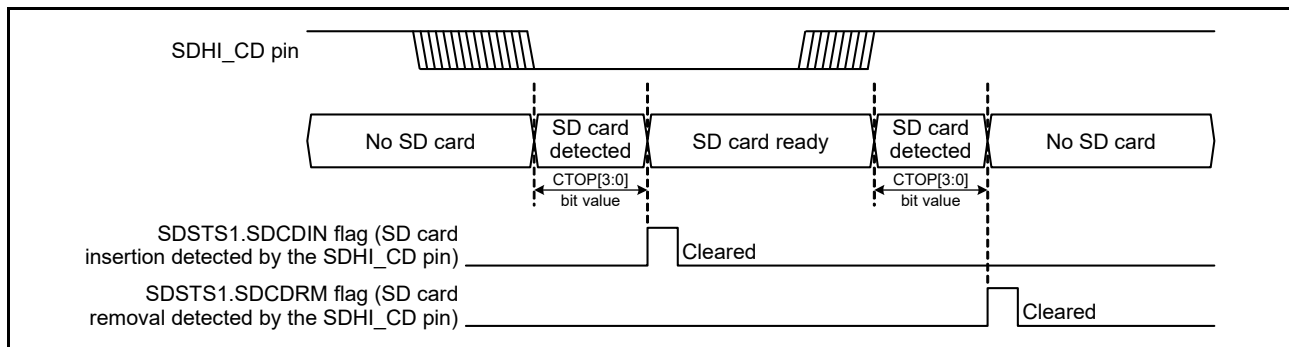


Figure 48.6 SD Card Detection Using the SDHI\_CD Pin

#### 48.3.3.2 Using the SDHI\_D3 Pin to Detect an SD Card

Figure 48.7 shows the timing chart for SD card detection using the SDHI\_D3 pin. The SDHI\_D3 pin is pulled-down by the MCU. The pull-down resistance value is determined by the specifications of the host device.

- Detecting SD card insertion

The signal from the SDHI\_D3 pin becomes high when an SD card is inserted. This causes the SDSTS1.SDD3IN flag to become 1. The SDSTS1.SDD3IN flag is cleared by setting it to 0.

- Detecting SD card removal

The signal from the SDHI\_D3 pin becomes low when the SD card is removed. This causes the SDSTS1.SDD3RM flag to become 1. The SDSTS1.SDD3RM flag is cleared by setting it to 0.

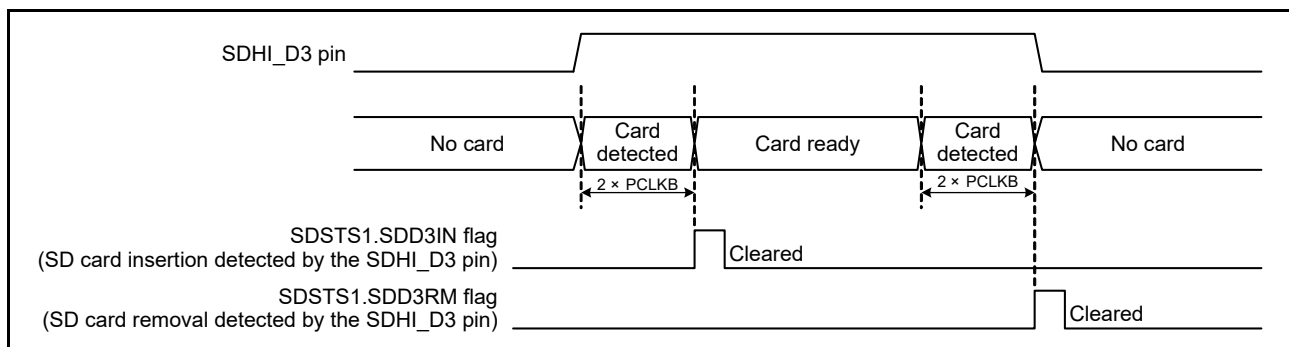


Figure 48.7 SD Card Detection Using the SDHI\_D3 Pin

### 48.3.4 SD Card Write Protection

The SDHI can disable writing to an SD card via the SDHI\_WP pin or a command.

#### 48.3.4.1 Using the SDHI\_WP Pin to Enable Write Protection

The SDHI\_WP pin is connected to the WP detection switch of the SD card connector, and the SDHI\_WP pin is pulled-down or pulled-up when an SD card is inserted. The resistance value and whether the SDHI\_WP pin is pulled-up or pulled-down are determined by the specifications of the host device. The status of the SDHI\_WP pin is reflected in the SDSTS1.SDWPMON flag. After an SD card is inserted, read the SDSTS1.SDWPMON flag to check if write protection is enabled or disabled.

#### 48.3.4.2 Using a Command to Enable Write Protection

The SDHI uses the write protect command or the SD card lock command to disable writing to the SD card.



### 48.3.5 Communication Errors and Timeouts

When a communication error or timeout error occurs, depending on the type of error, the corresponding status flag in the SDSTS2 register becomes 1. Also, depending on the source of the error, the corresponding flag in the SDERSTS1 or SDERSTS2 register becomes 1.

The status flags in registers SDERSTS1 and SDERSTS2 become 0 by writing to the SDCMD register, or by setting the SDRST.SDRST bit to 0.

**Table 48.6 Communication Errors**

| Communication Error | Interrupt Flag Register |            | Error Status Register |            | This Occurs When...                                                                             |
|---------------------|-------------------------|------------|-----------------------|------------|-------------------------------------------------------------------------------------------------|
|                     | Register symbol         | Bit symbol | Register symbol       | Bit symbol |                                                                                                 |
| End bit error       | SDSTS2                  | ENDE       | SDERSTS1              | CRCLENE    | The CRC status token length is in error                                                         |
|                     |                         |            |                       | RDLENE     | The read data length is in error                                                                |
|                     |                         |            |                       | RSPLENE1   | The response length is in error *1                                                              |
|                     |                         |            |                       | RSPLENE0   | The response length is in error *2                                                              |
| CRC error           |                         | CRCE       |                       | CRCTKE     | The CRC status token is in error                                                                |
|                     |                         |            |                       | RDCRCE     | There is a CRC error in the read data                                                           |
|                     |                         |            |                       | RSPCRCE1   | There is a CRC error in the response *1                                                         |
|                     |                         |            |                       | RSPCRCE0   | There is a CRC error in the response *2                                                         |
| Command error       |                         | CMDE       |                       | CMDE1      | The command index field value for the transmitted command and received response do not match *1 |
|                     |                         |            |                       | CMDE0      | The command index field value for the transmitted command and received response do not match *2 |

Note 1. CMD12 or CMD52 which are automatically issued to stop transfer.

Note 2. A command other than CMD12 or CMD52 which are automatically issued to stop transfer.

**Table 48.7 Timeouts**

| Timeout                                   | Interrupt Flag Register |                                                                                                                                                        | Error Status Register |            | This Occurs When...                                                                                                              |
|-------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------|----------------------------------------------------------------------------------------------------------------------------------|
|                                           | Register symbol         | Bit symbol                                                                                                                                             | Register symbol       | Bit symbol |                                                                                                                                  |
| Response timeout                          | SDSTS2                  | RSPTO                                                                                                                                                  | SDERSTS2              | RSPTO1     | A response is not received even after a minimum of 640 SDHI clock cycles elapse *1                                               |
|                                           |                         |                                                                                                                                                        |                       | RSPTO0     | A response is not received even after a minimum of 640 SDHI clock cycles elapse *2                                               |
| Data timeout (excluding response timeout) |                         | DTO                                                                                                                                                    |                       | CRCBSYTO   | After the CRC status token is received, the SDHI is busy for at least the period set *3                                          |
|                                           |                         |                                                                                                                                                        |                       | CRCTO      | After the write data is transmitted, the CRC status token is not received even after at least the period set *3 elapses          |
|                                           |                         |                                                                                                                                                        |                       | RDTO       | After the read command is issued, the read data is not received even after at least the period set *3 elapses                    |
|                                           |                         |                                                                                                                                                        |                       |            | After the read data is received, the next block read data is not received even after at least the period set *3 elapses          |
|                                           |                         |                                                                                                                                                        |                       |            | After the SDHI exits the read wait state, the next block read data is not received even after at least the period set *3 elapses |
|                                           |                         |                                                                                                                                                        |                       | BSYTO1     | After CMD12 is issued during the command sequence, the SDHI is busy for at least the period set *3                               |
| BSYTO0                                    |                         | After the R1b response is received, the SDHI is busy for at least the period set *3 (a command other than CMD12 is issued during the command sequence) |                       |            |                                                                                                                                  |

Note 1. CMD12 or CMD52 which are automatically issued to stop transfer.

Note 2. A command other than CMD12 or CMD52 which are automatically issued to stop transfer.

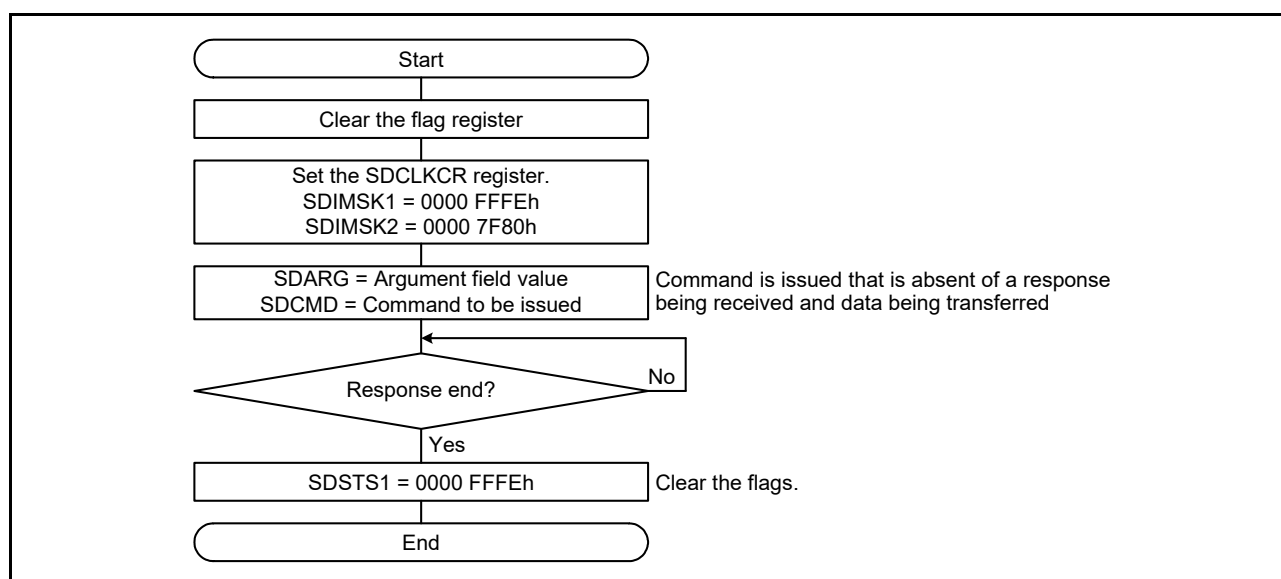
Note 3. The period is set in the SDOPT.TOP[3:0] bits.

### 48.3.6 Examples of Issuing a Command

#### 48.3.6.1 Command Absent of Response Reception and Data Transfer

Figure 48.8 shows an example of no response being received and no data being transferred after the SDHI issues a command.

1. Set the flags in registers SDSTS1 and SDSTS2 to 0.
2. Set the SDHI clock in the SDCLKCR register, and set the interrupt requests to be masked in registers SDIMSK1 and SDIMSK2. Refer to section 48.5.5 for details on setting the SDCLKCR register.
3. After setting the argument field value to the SDARG register, write the command information to be sent to the SDCMD register. The SDHI issues a command when a value is written to the SDCMD register.
4. After a command is issued, the SDSTS1.RSPEND flag becomes 1, and a response end interrupt request is generated.
5. Set the SDSTS1.RSPEND flag to 0.



**Figure 48.8** Command Issued That Is Absent of Response Reception and Data Transfer

### 48.3.6.2 Command Absent of Data Transfer

Figure 48.9 shows an example of no data being transferred after the SDHI issues a command.

1. Set the flags in registers SDSTS1 and SDSTS2 to 0.
2. Set the SDHI clock in the SDCLKCR register, and set the interrupt requests to be masked in registers SDIMSK1 and SDIMSK2. Refer to section 48.5.5 for details on setting the SDCLKCR register.
3. After setting the argument field value in the SDARG register, write the information of the command to be issued to the SDCMD register. The SDHI issues a command when a value is written to the SDCMD register.
4. After a response is received, the SDSTS1.RSPEND flag becomes 1, and a response end interrupt request is generated.
5. Set the SDSTS1.RSPEND flag to 0, and read the response stored in the SDRSP10 register.

Perform error processing (clear the interrupt flag) if a communication error or timeout occurs.

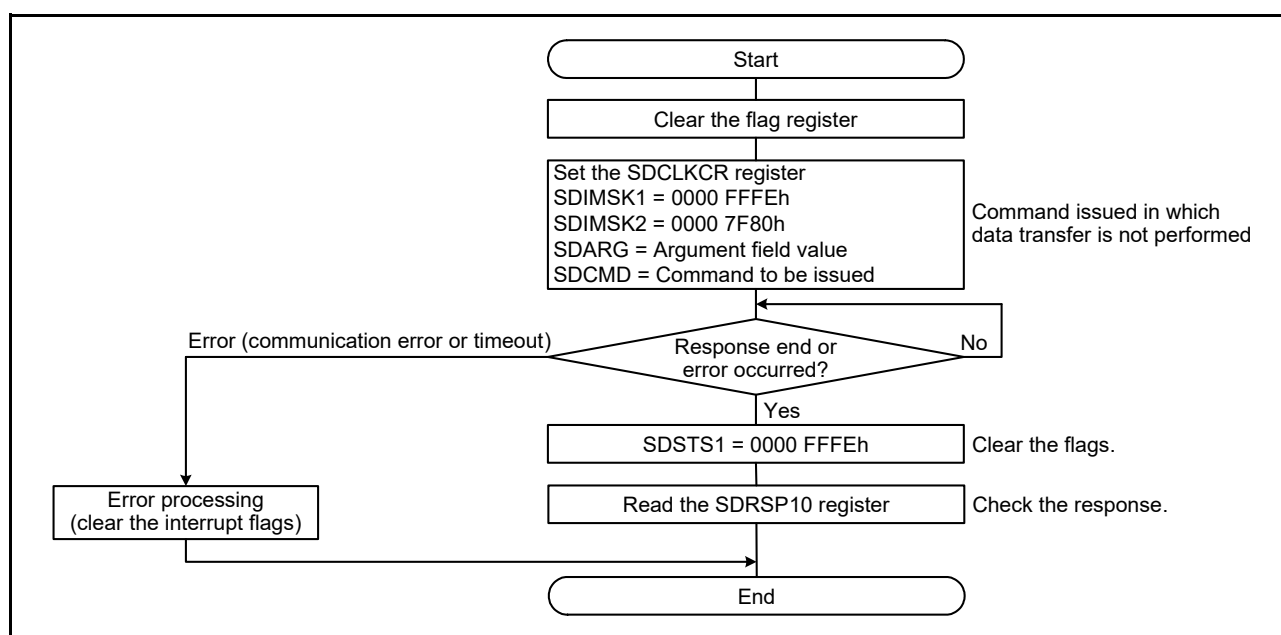


Figure 48.9 Command Issued That Is Absent of Data Transfer

### 48.3.6.3 Single Block Read Command (CMD17)

Figure 48.10 shows an example of issuing the single block read command (CMD17).

1. Set the flags in registers SDSTS1 and SDSTS2 to 0.
2. Set the SDHI clock in the SDCLKCR register, and set the interrupt requests to be masked in registers SDIMSK1 and SDIMSK2. Refer to **section 48.5.5** for details on setting the SDCLKCR register.
3. After setting the argument field value for CMD17 to the SDARG register, write 0000 0011h to the SDCMD register. The SDHI issues CMD17 when a value is written to the SDCMD register.
4. When the response is received, the SDSTS1.RSPEND flag becomes 1, and a response end interrupt request is generated.
5. Set the SDSTS1.RSPEND flag to 0, and read the response stored in the SDRSP10 register. If the response read is in error, set the SDSTOP.STP bit or SDIOMD.IOABT bit to 1, and the command sequence can be stopped. When the command sequence is stopped, the SDSTS1.ACEND flag becomes 1. Note that CMD12 and CMD52 are not automatically issued by stopping this command sequence.
6. After the response is received, set the SDIMSK1.ACENDM bit to 0 and set the SDIMSK2.BREM bit to 0.
7. After the amount of data set in the SDSIZE.LEN[9:0] bits is received from the SD card, the SDSTS2.BRE flag becomes 1, and the BRE interrupt request is generated.
8. Set the SDSTS2.BRE flag to 0, and read the amount of data set in the SDSIZE.LEN[9:0] bits from the SDBUFR register.
9. After data has been read from the SDBUFR register, the SDSTS1.ACEND flag becomes 1, and the access end interrupt request is generated.
10. Set the SDSTS1.ACEND flag to 0.

Perform error processing (clear the interrupt flag) if a communication error or timeout occurs.

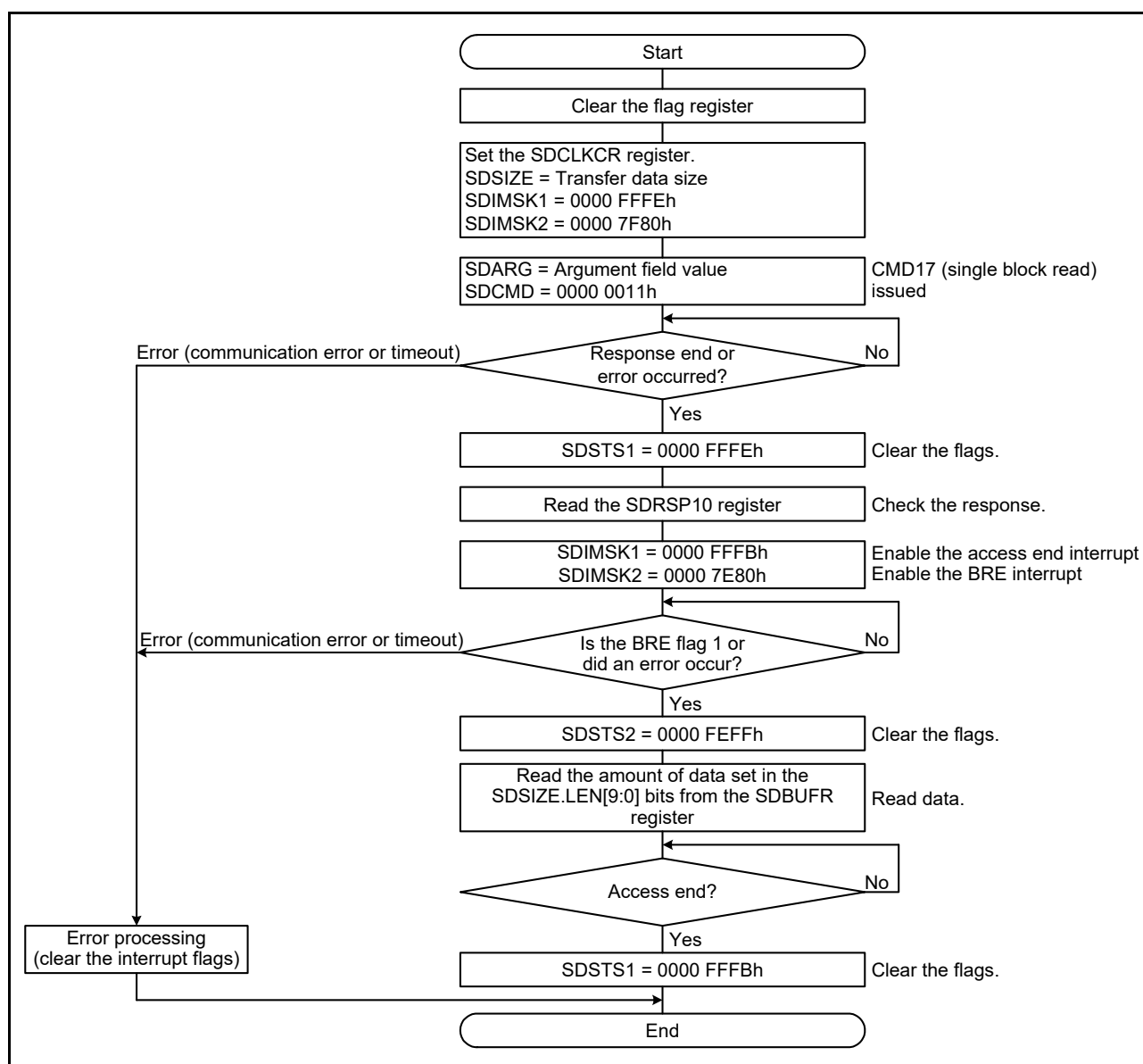


Figure 48.10 Issuing the Single Block Read Command

#### 48.3.6.4 Single Block Write Command (CMD24)

Figure 48.11 shows an example of issuing the single block write command (CMD24).

1. Set the flags in registers SDSTS1 and SDSTS2 to 0.
2. Set the SDHI clock in the SDCLKCR register, and set the interrupt requests to be masked in registers SDIMSK1 and SDIMSK2. Refer to **section 48.5.5** for details on setting the SDCLKCR register.
3. After setting the argument field value for CMD24 to the SDARG register, write 0000 0018h to the SDCMD register. The SDHI issues CMD24 when a value is written to the SDCMD register.
4. When the response is received, the SDSTS1.RSPEND flag becomes 1, and the response end interrupt request is generated.
5. Set the SDSTS1.RSPEND flag to 0 and read the response stored in the SDRSP10 register. If the read response is in error, set the SDSTOP.STP bit or SDIOMD.IOABT bit to 1, and the command sequence can be stopped. When the command sequence is stopped, the SDSTS1.ACEND flag becomes 1. Note that when this command sequence is stopped, CMD12 and CMD52 are not automatically issued.
6. After the response is received, set the SDIMSK1.ACENDM bit to 0, and set the SDIMSK2.BWEM bit to 0.
7. When the SDBUFR register becomes write accessible, the SDSTS2.BWE flag becomes 1, and the BWE interrupt request is generated.
8. Set the SDSTS2.BWE flag to 0, and write the amount of data set in the SDSIZE.LEN[9:0] bits to the SDBUFR register. After writing to the SDBUFR register, the SDHI transmits write data to the SD card. Also, after writing to the SDBUFR register, data transmission may cause a communication error or timeout to occur.
9. After all data has been written to the SD card, the SDHI receives the CRC status token, and the SDHI\_D0 pin line becomes busy (low). Then, when the SDHI exits the busy state, the SDSTS1.ACEND flag becomes 1, and the access end interrupt request is generated.
10. Set the SDSTS1.ACEND flag to 0.

Perform error processing (clear the interrupt flag) if a communication error or timeout occurs.

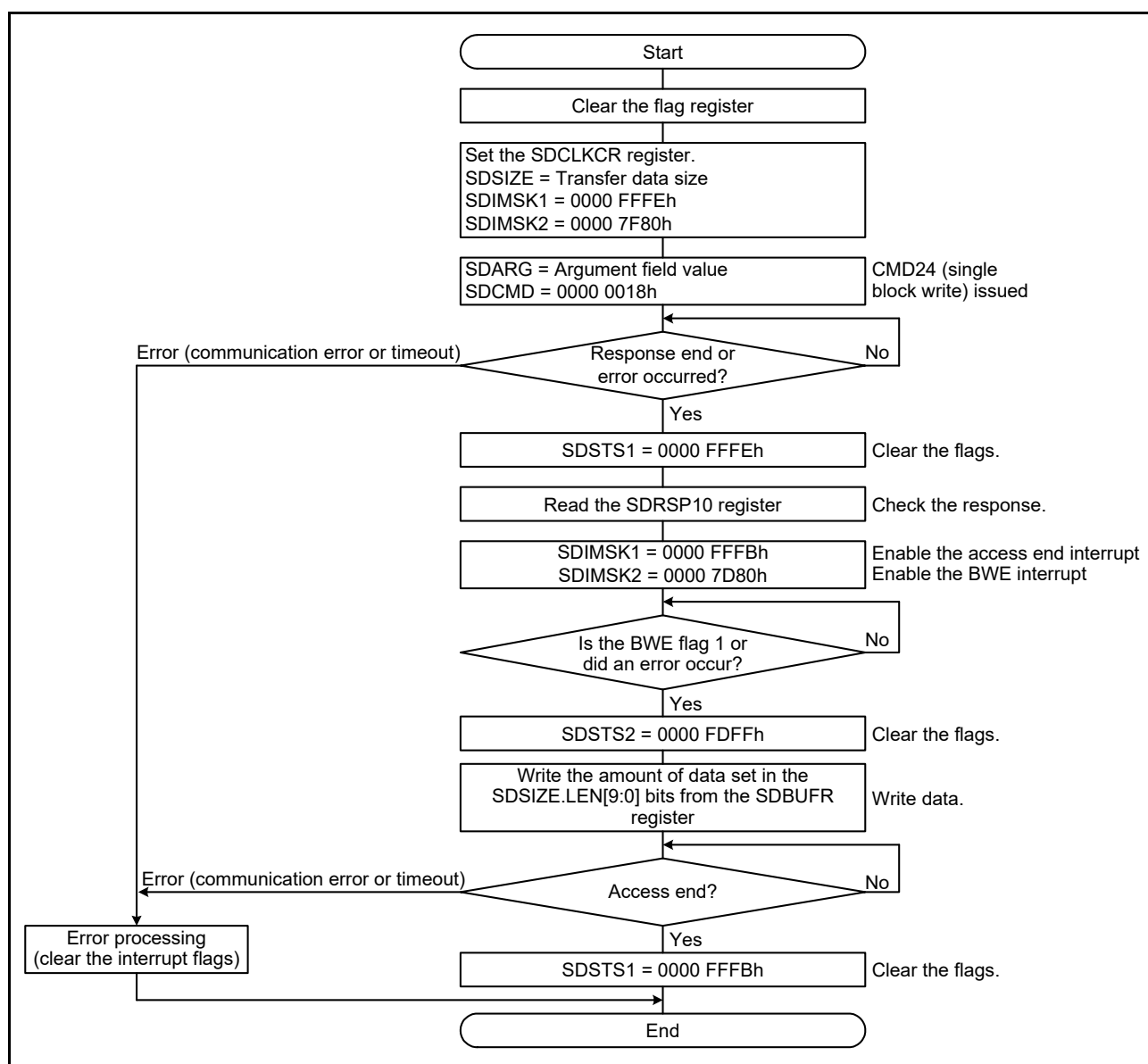


Figure 48.11 Issuing the Single Block Write Command

### 48.3.6.5 Multi-Block Read Command (CMD18)

Figure 48.12 shows an example of issuing the multi-block read command (CMD18).

1. Set the flags in registers SDSTS1 and SDSTS2 to 0.
2. Set the SDHI clock in the SDCLKCR register, and set the interrupt requests to be masked in registers SDIMSK1 and SDIMSK2. Refer to [section 48.5.5](#) for details on setting the SDCLKCR register. Set the SDSTOP.SDBLKCNTEN bit to 1, and set the number of transfer blocks in the SDBLKCNT register.
3. After setting the argument field value for CMD18 to the SDARG register, write 0000 0012h to the SDCMD register. The SDHI issues CMD18 when a value is written to the SDCMD register.
4. When the response is received, the SDSTS1.RSPEND flag becomes 1, and the response end interrupt request is generated.
5. Set the SDSTS1.RSPEND flag to 0 and read the response stored in the SDRSP54 register. If the read response is in error, set the SDSTOP.STP bit to 1, and the command sequence can be stopped. When the SDSTOP.STP bit is set to 1, the SDHI automatically issues CMD12, and the response is received. At this point, the SDSTS1.ACEND flag becomes 1, and if the access end interrupt request is enabled, the access end interrupt request is generated. Next, set the SDSTS1.ACEND flag to 0 and read the response.
6. After the response is received, set the SDIMSK1.ACENDM bit to 0, and set the SDIMSK2.BREM bit to 0.
7. After receiving one block of data from the SD card, the SDSTS2.BRE bit becomes 1, and the BRE interrupt request is generated.
8. Set the SDSTS2.BRE flag to 0, and read the amount of data set in the SDSIZE.LEN[9:0] bits from the SDBUFR register. The read access to the SDBUFR register repeats for the amount of transfer blocks set in the SDBLKCNT register. Also, while reading the SDBUFR register, data reception may cause a communication error or timeout to occur. After the amount of transfer blocks set in the SDBLKCNT register have been read, the SDHI automatically issues CMD12, and the response is received. At this time, the SDHI automatically writes 0000 0000h to the SDARG register.
9. When all blocks have been received and the CMD12 response is received, the SDSTS1.ACEND flag becomes 1 and the access end interrupt request is generated.
10. Set the SDSTS1.ACEND flag to 0 and read the response.

Perform error processing (clear the interrupt flag) if a communication error or timeout occurs.



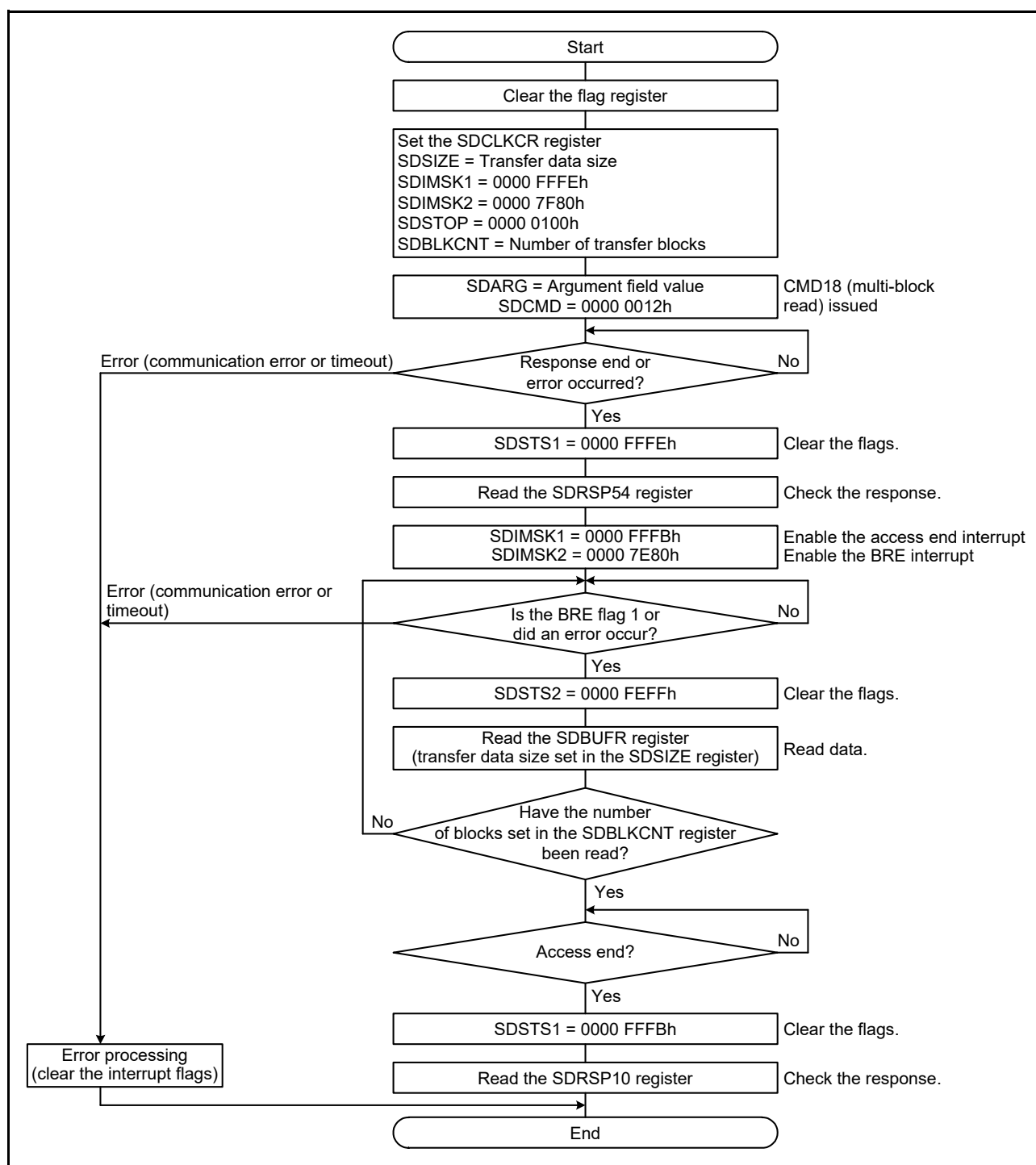


Figure 48.12 Issuing the Multi-Block Read Command

### 48.3.6.6 Multi-Block Write Command (CMD25)

Figure 48.13 shows an example of issuing the multi-block write command (CMD25).

1. Set the flags in registers SDSTS1 and SDSTS2 to 0.
2. Set the SDHI clock in the SDCLKCR register, and set the interrupt requests to be masked in registers SDIMSK1 and SDIMSK2. Refer to **section 48.5.5** for details on setting the SDCLKCR register. Set the SDSTOP.SDBLKCNTEN bit to 1, and set the number of transfer blocks in the SDBLKCNT register.
3. After setting the argument field value for CMD25 to the SDARG register, write 0000 0019h to the SDCMD register. The SDHI issues CMD25 when a value is written to the SDCMD register.
4. When the response is received, the SDSTS1.RSPEND flag becomes 1, and the response end interrupt request is generated.
5. Set the SDSTS1.RSPEND flag to 0 and read the response stored in the SDRSP54 register. If the read response is in error, set the SDSTOP.STP bit to 1, and the command sequence can be stopped. When the SDSTOP.STP bit is set to 1, the SDHI automatically issues CMD12, and the response is received. At this point, the SDSTS1.ACEND flag becomes 1, and if the access end interrupt request is enabled, the access end interrupt request is generated. Next, set the SDSTS1.ACEND flag to 0 and read the response.
6. After the response is received, configure the SDIMSK1 register to enable the access end interrupt request, and configure the SDIMSK2 register to enable the BWE interrupt request.
7. When the SDBUFR register becomes write accessible, the SDSTS2.BWE flag becomes 1, and the BWE interrupt request is generated.
8. Set the SDSTS2.BWE flag to 0, and write the amount of data set in the SDSIZE.LEN[9:0] bits to the SDBUFR register. After writing to the SDBUFR register, and after the SDHI transmits write data to the SD card, the CRC status token is received, and the SDHI\_D0 pin line becomes busy (low). The write access to the SDBUFR register and CRC status token reception repeat for the amount of transfer blocks set in the SDBLKCNT register. Also, after writing to the SDBUFR register, data transmission may cause a communication error or timeout to occur. After the amount of transfer blocks set in the SDBLKCNT register have been written, the SDHI automatically issues CMD12, and the response is received. At that time, the SDHI automatically writes 0000 0000h to the SDARG register.
9. When all blocks have been transmitted and the CRC status token is received, the SDHI exits the busy state, the SDSTS1.ACEND flag becomes 1, and the access end interrupt request is generated.
10. Set the SDSTS1.ACEND flag to 0 and read the response.

Perform error processing (clear the interrupt flag) if a communication error or timeout occurs.

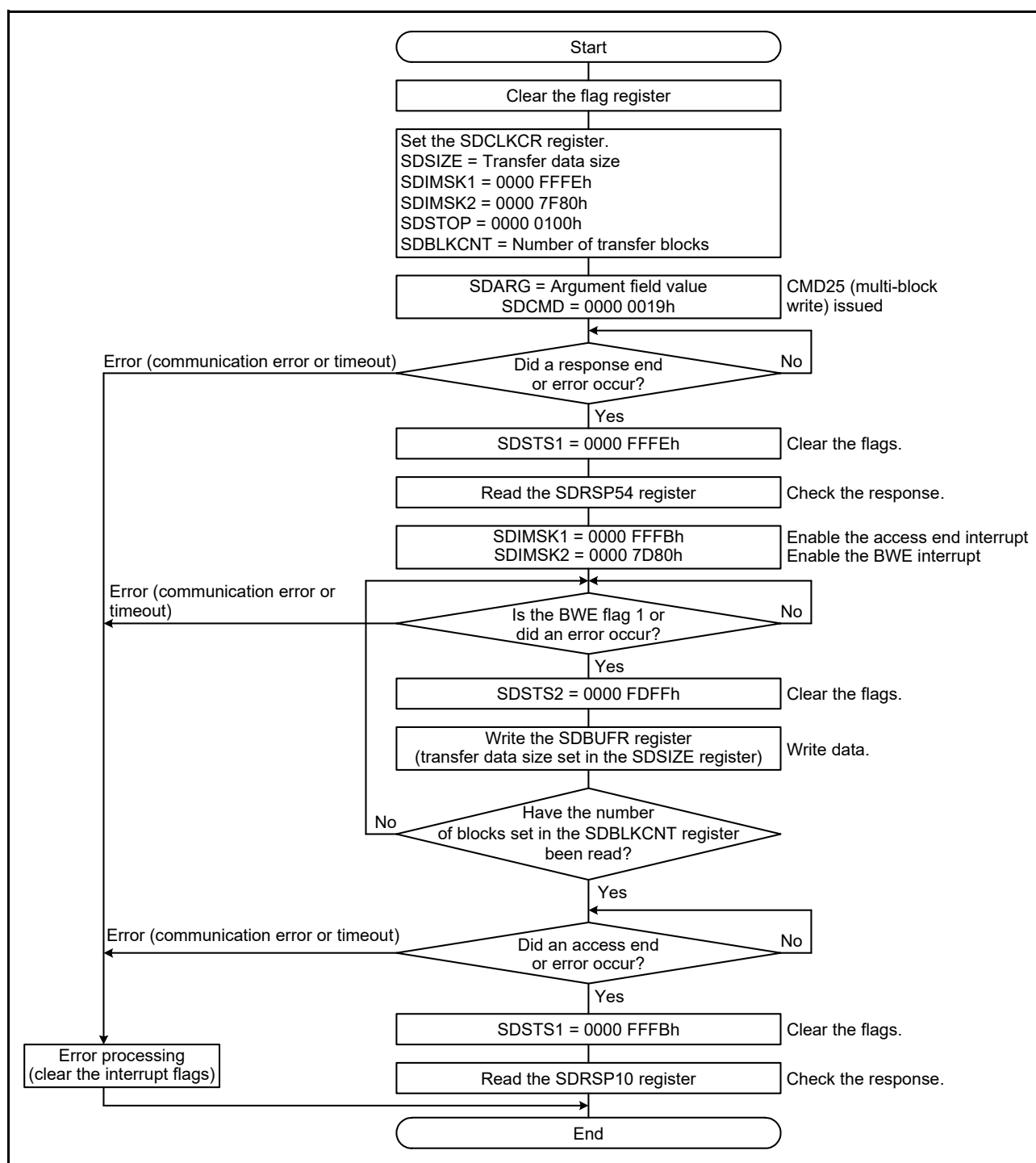


Figure 48.13 Issuing the Multi-Block Write Command

## 48.3.6.7 IO\_RW\_DIRECT Command (CMD52)

Figure 48.14 shows an example of issuing the IO\_RW\_DIRECT command (CMD52).

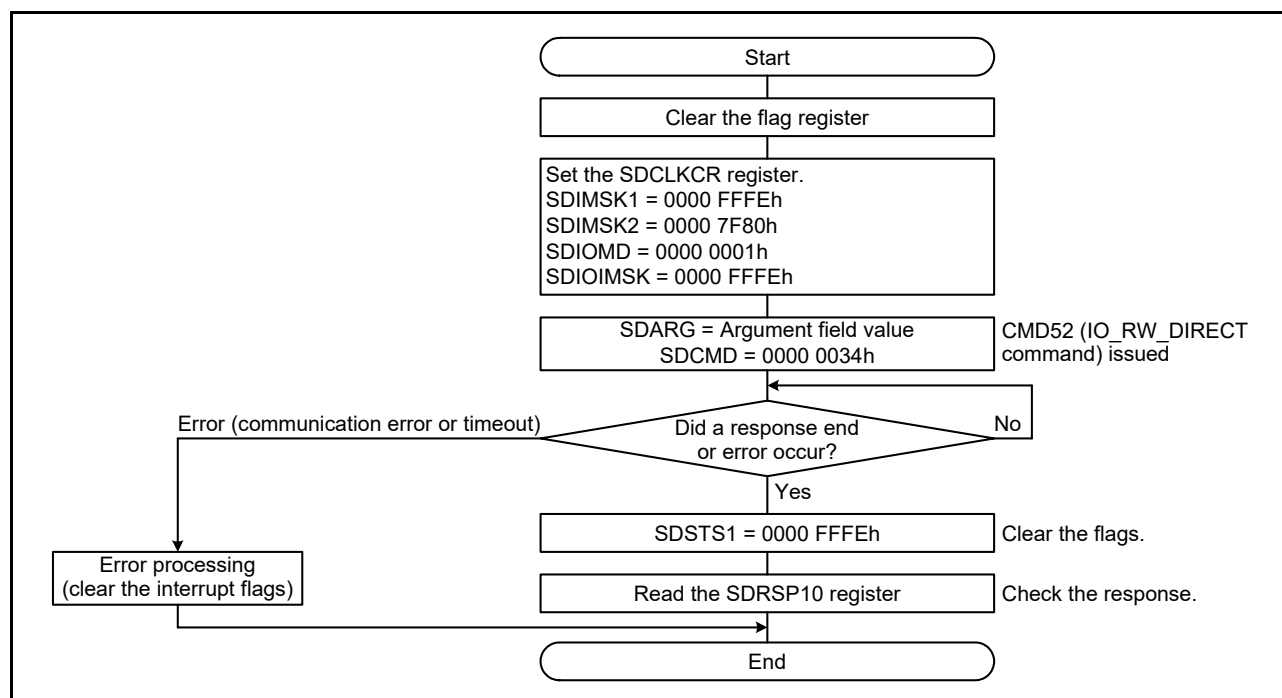


Figure 48.14 Issuing the IO\_RW\_DIRECT Command

## 48.3.6.8 IO\_RW\_EXTENDED Command (CMD53 (Multi-Block Read))

Figure 48.15 shows an example of issuing the IO\_RW\_EXTENDED command (CMD53/Multi-block read).

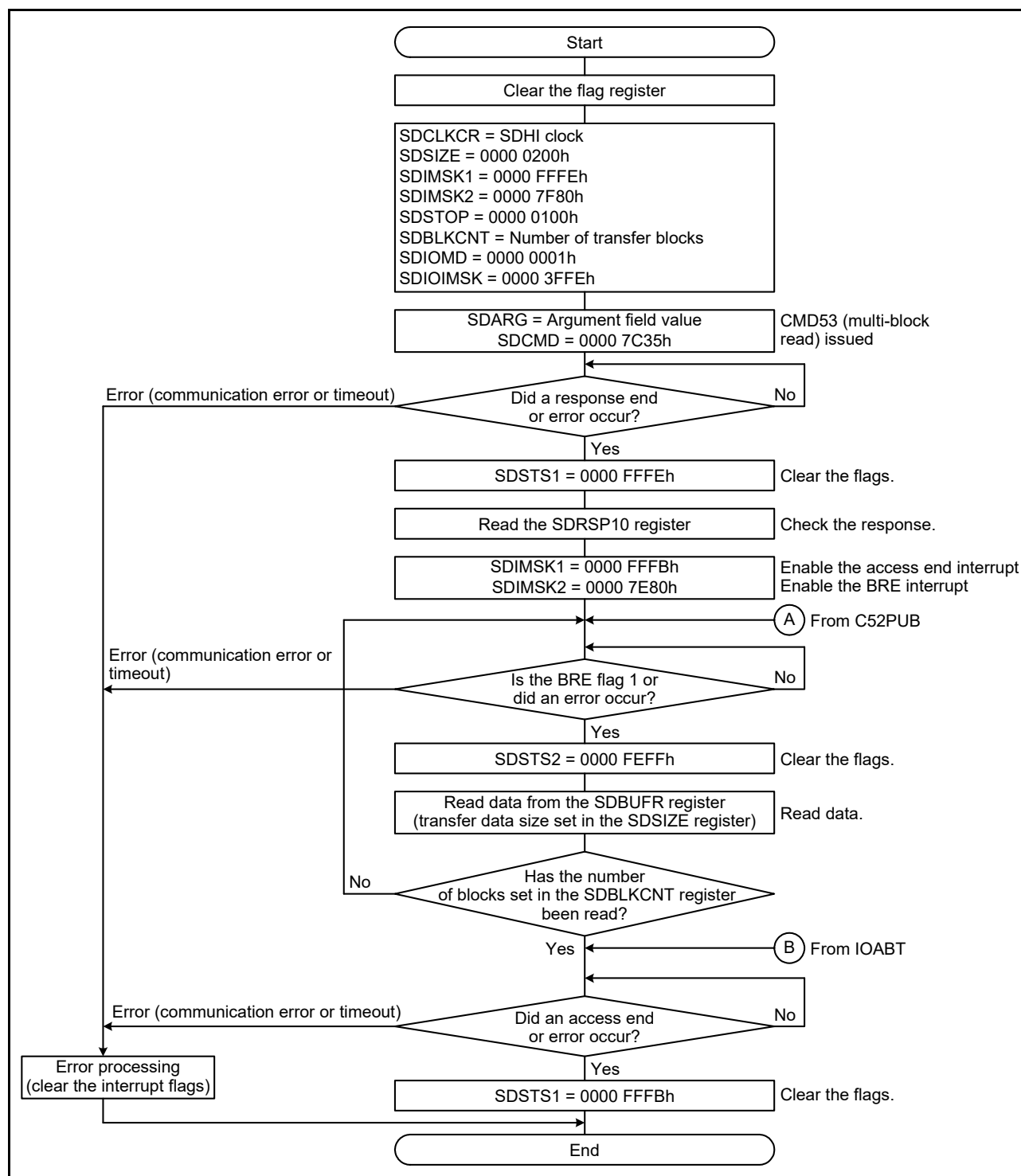
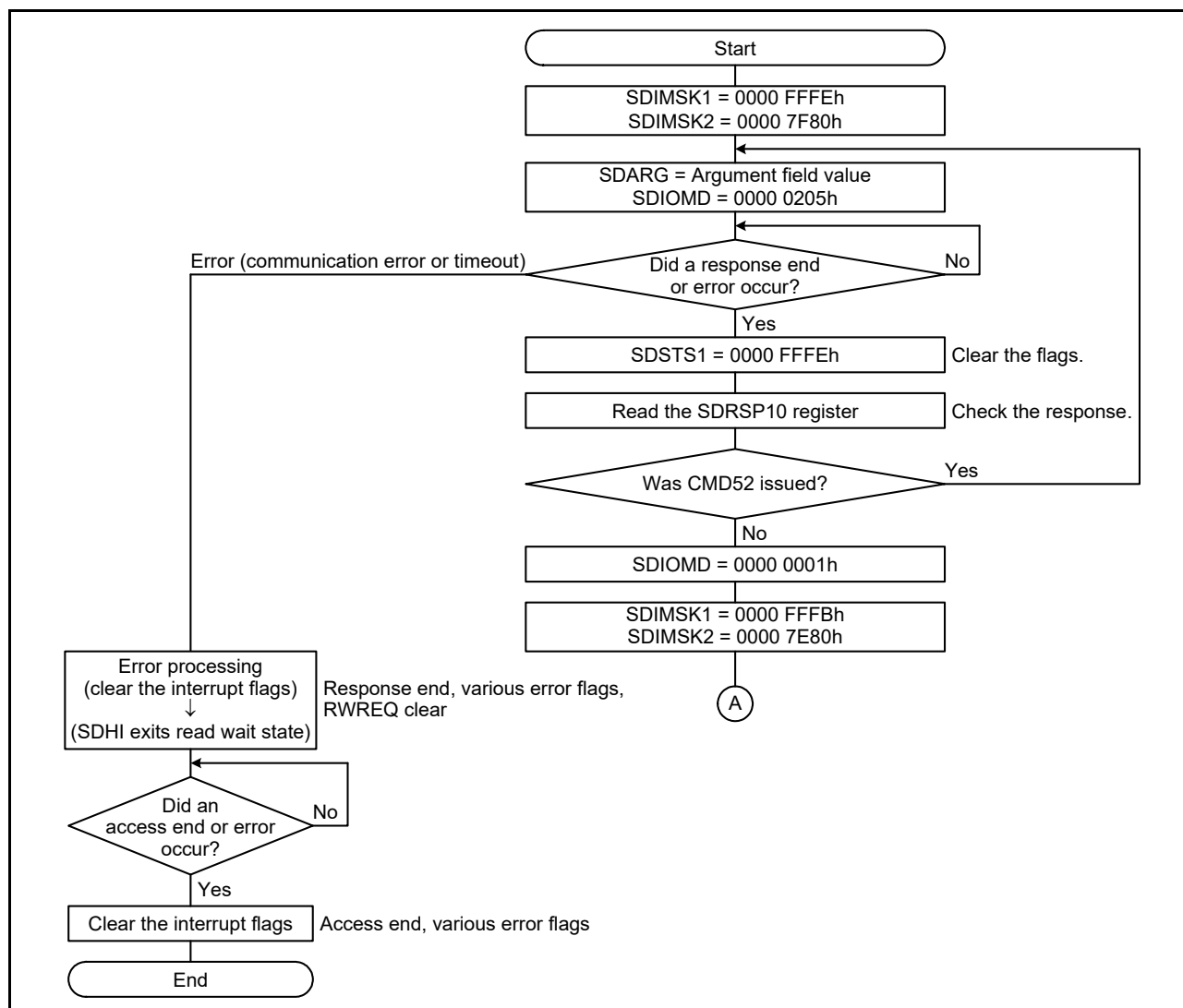


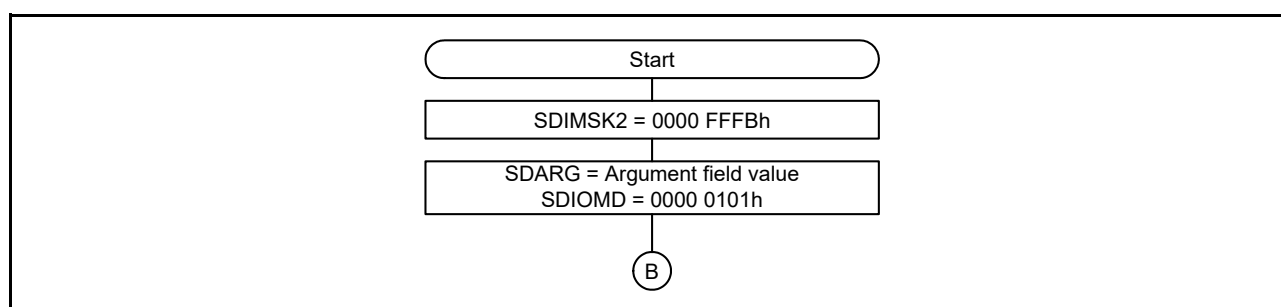
Figure 48.15 Issuing the IO\_RW\_EXTENDED Command (CMD53/Multi-Block Read)

Figure 48.16 shows an example of entering the read wait state and then issuing the SDIO none abort command (CMD52) during the IO\_RW\_EXTENDED command (CMD53/Multi-block read).



**Figure 48.16** Entering the Read Wait State and Then Issuing the SDIO None Abort Command (CMD52) During the IO\_RW\_EXTENDED Command (CMD53/Multi-Block Read)

Figure 48.17 shows an example of SDIO abort (CMD52) issued during IO\_RW\_EXTENDED command (CMD53/Multi-block read) sequence.



**Figure 48.17** SDIO Abort (CMD52) Issued During IO\_RW\_EXTENDED Command (CMD53/Multi-Block Read) Sequence

## 48.3.6.9 IO\_RW\_EXTENDED (CMD53 Multi-Block Write)

Figure 48.18 shows an example of issuing the IO\_RW\_EXTENDED command (CMD53/Multi-block write).

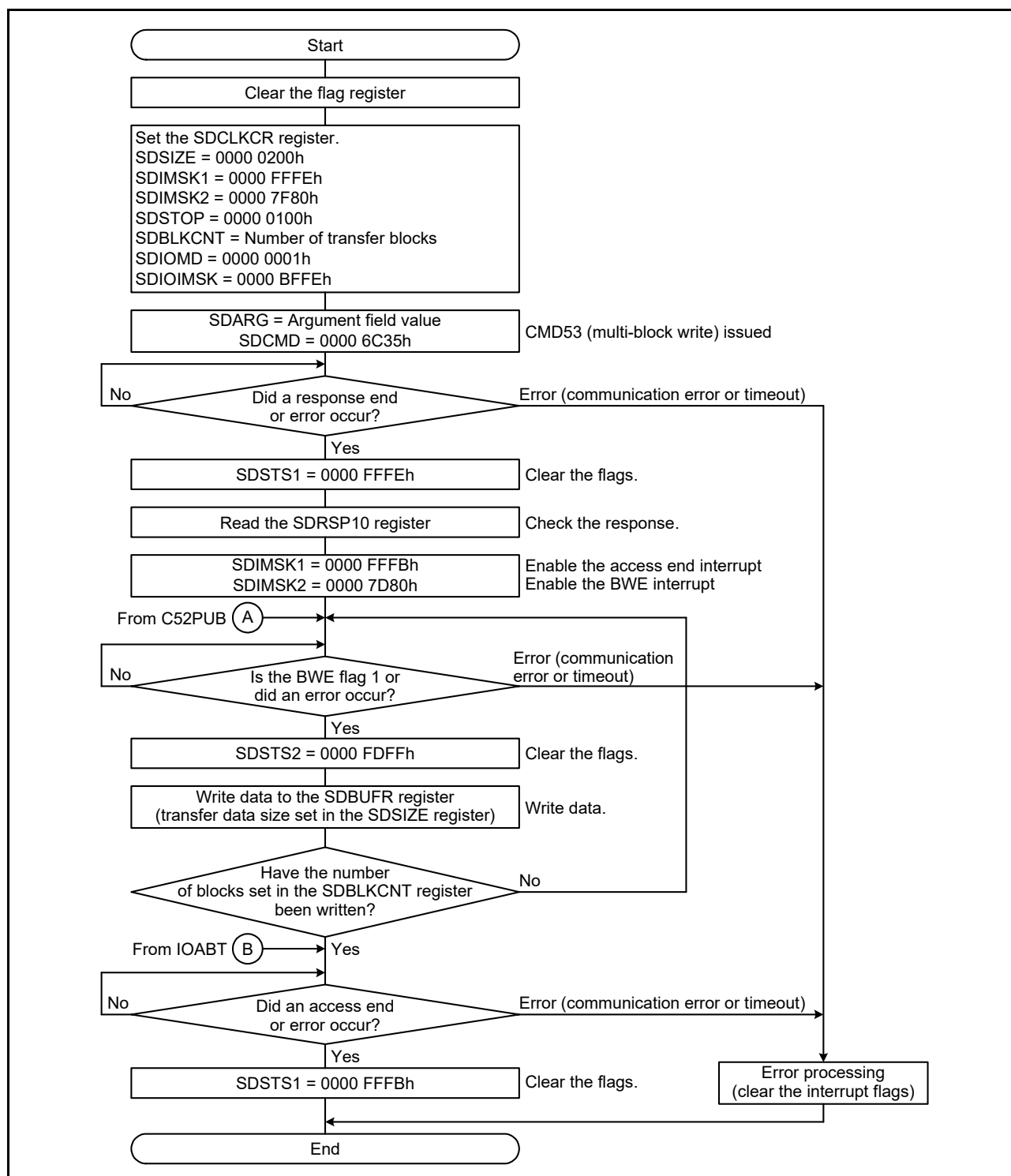
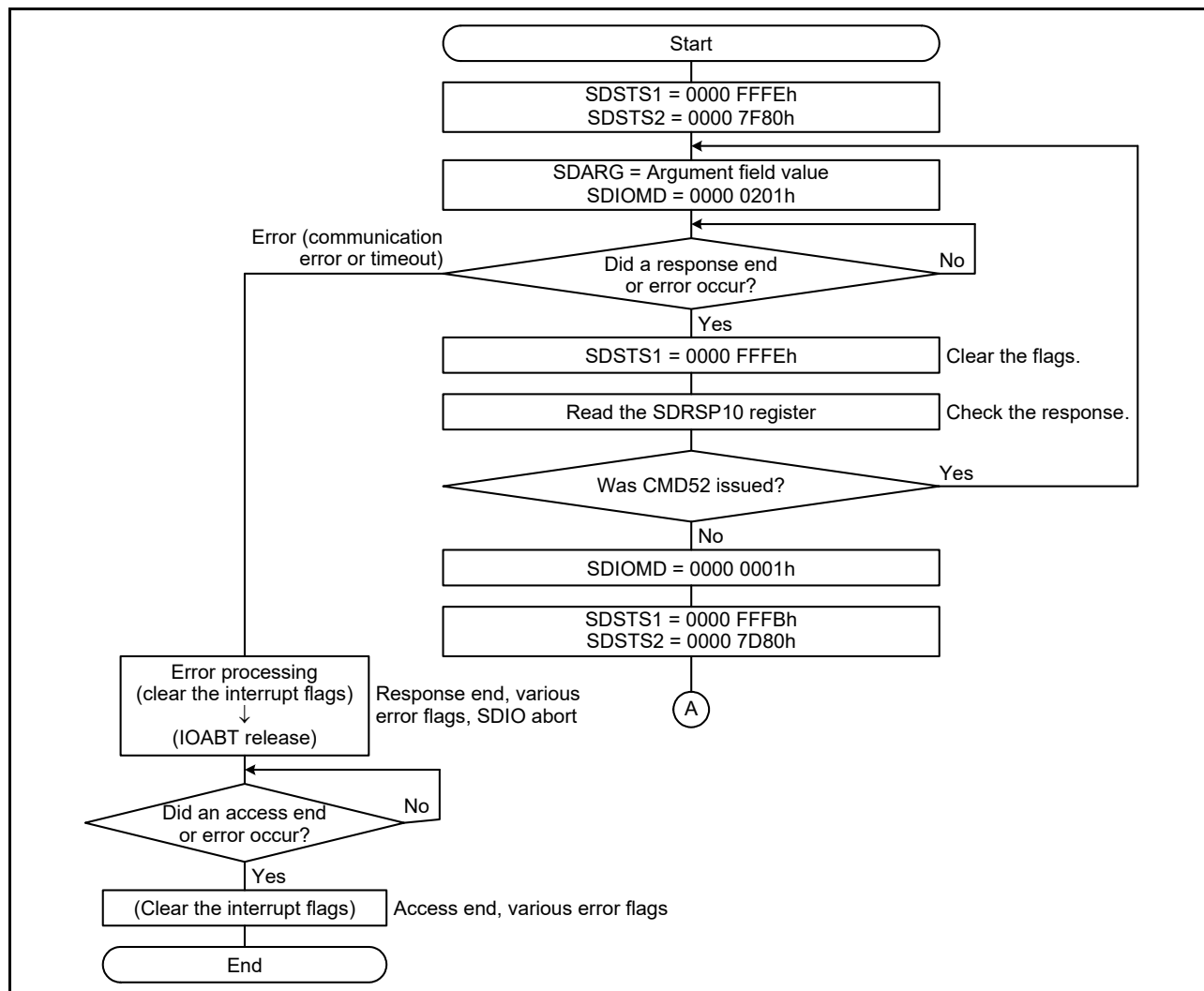


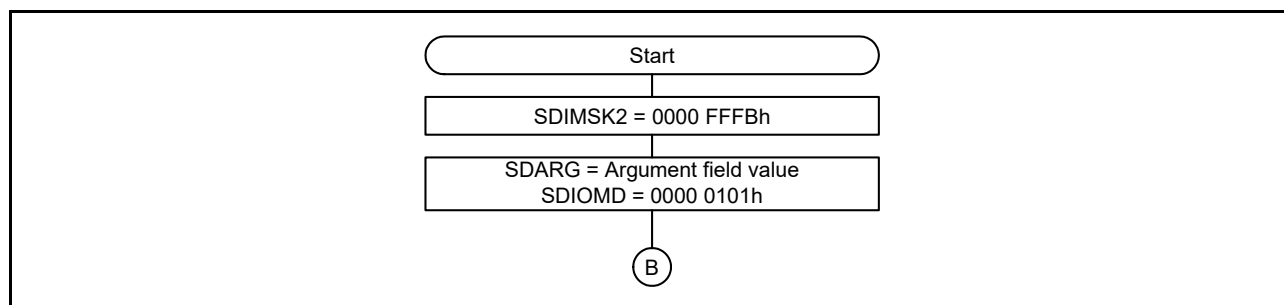
Figure 48.18 Issuing the IO\_RW\_EXTENDED Command (CMD53/Multi-Block Write)

Figure 48.19 shows SDIO none abort (CMD52) issued during IO\_RW\_EXTENDED command (CMD53/Multi-block write) sequence.



**Figure 48.19 SDIO None Abort (CMD52) Issued During IO\_RW\_EXTENDED Command (CMD53/Multi-Block Write) Sequence**

Figure 48.20 shows SDIO abort (CMD52) issued during IO\_RW\_EXTENDED command (CMD53) sequence.



**Figure 48.20 SDIO Abort (CMD52) Issued During IO\_RW\_EXTENDED Command (CMD53) Sequence**



## 48.3.6.10 DMA Transfer

Figure 48.21 shows an example of data being transferred from the SDBUFR register after the CMD18 multi-block read command is issued.

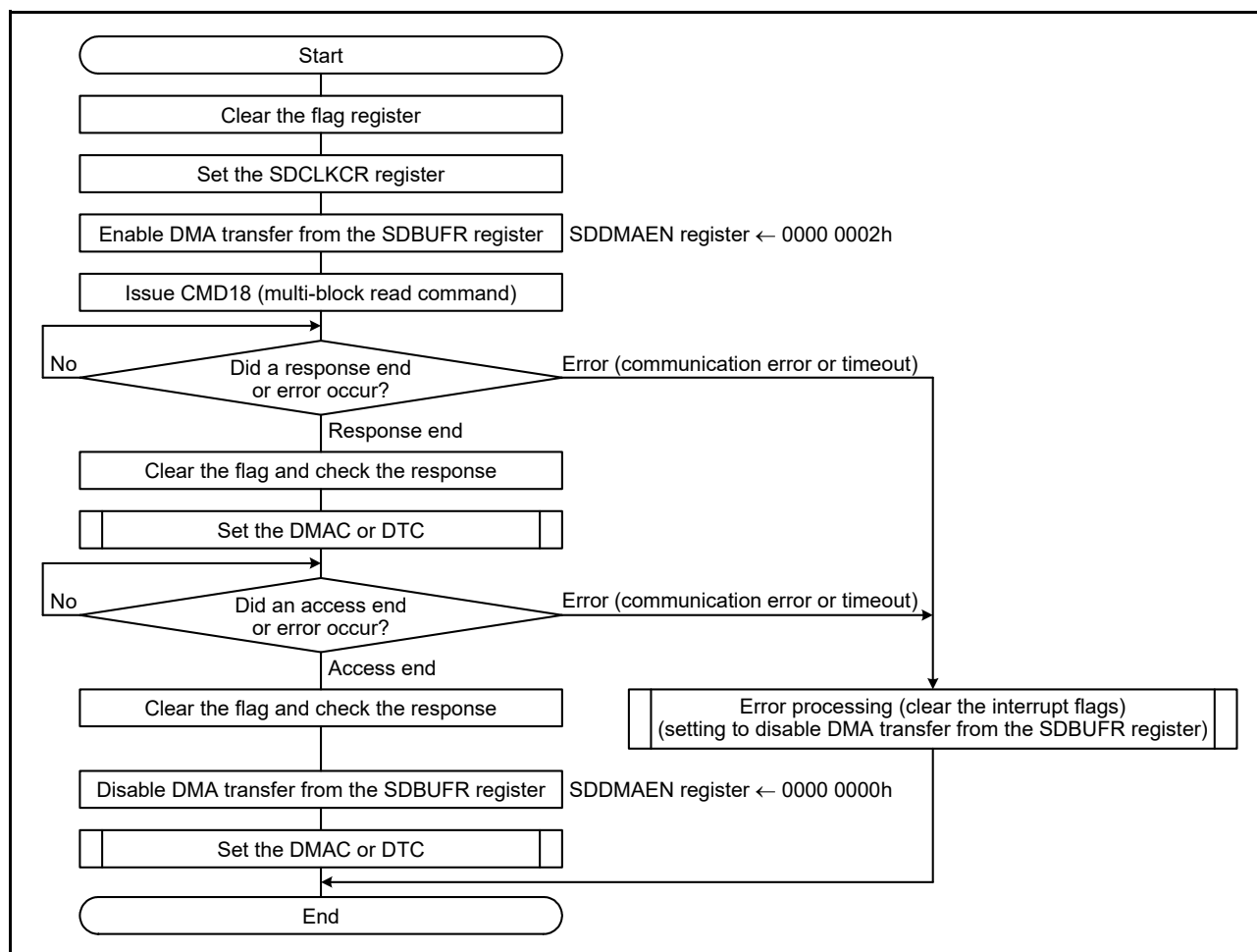


Figure 48.21 DMA Transfer After CMD18 is Issued

Figure 48.22 shows an example of data being DMA transferred to the SDBUFR register after the CMD25 multi-block write command is issued.

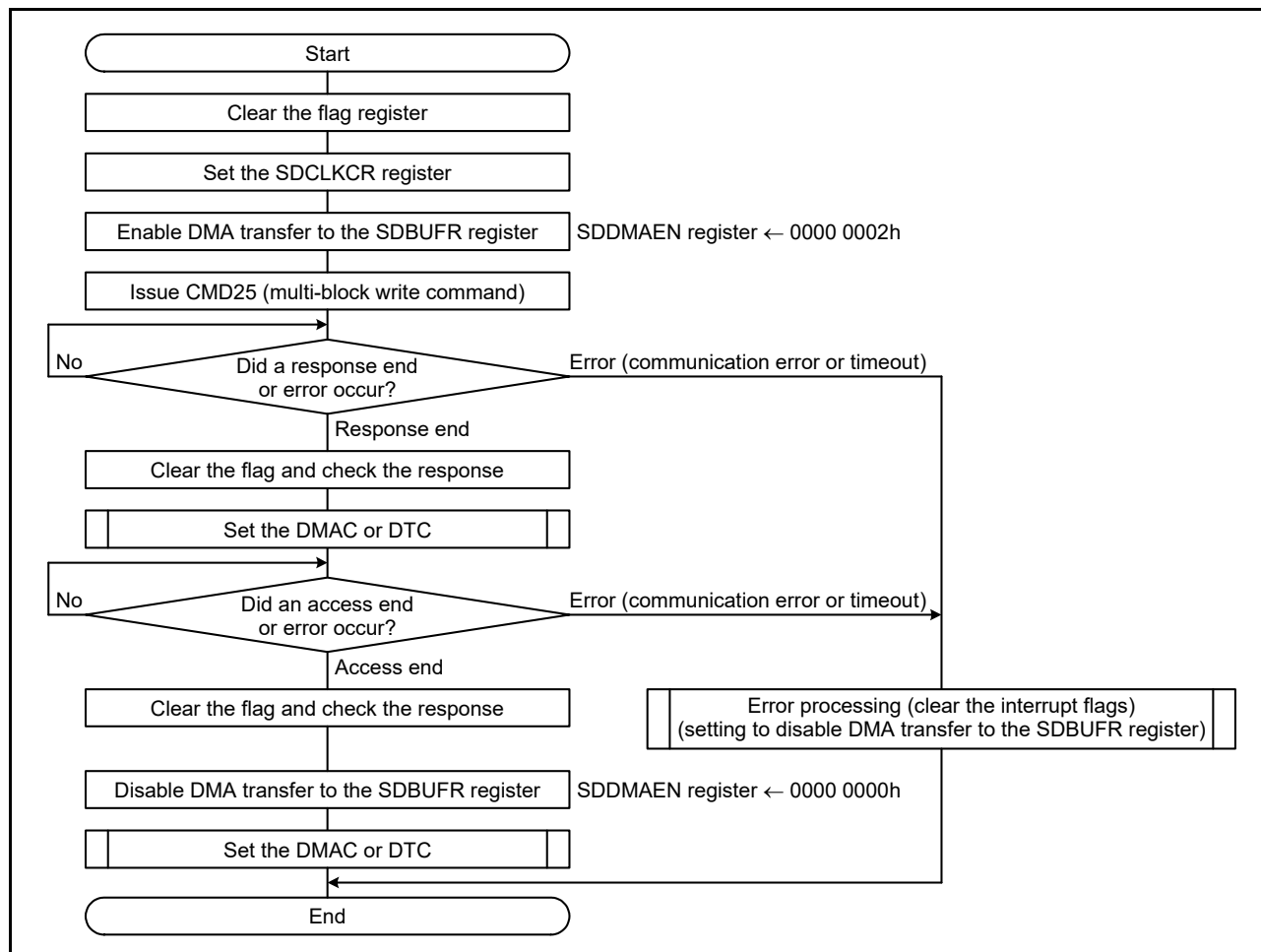


Figure 48.22 DMA Transfer After CMD25 is Issued

## 48.4 Interrupts

Table 48.8 lists the SDHI interrupt sources. When the status flags in registers SDSTS1, SDSTS2, and SDIOSTS become 1, if the corresponding bits in registers SDIMSK1, SDIMSK2, and SDIOIMSK are 0, the SDHI requests an interrupt. When clearing the status flags in registers SDSTS1, SDSTS2, and SDIOSTS, write 0 to the status flags to be cleared and write 1 to the states flags not being cleared.

**Table 48.8 Interrupt Sources**

| Interrupt Source | Status Flag Register |            | Interrupt Mask/Enable Register |            | DMAC/DTC Triggerable |
|------------------|----------------------|------------|--------------------------------|------------|----------------------|
|                  | Register symbol      | Bit symbol | Register symbol                | Bit symbol |                      |
| CACI             | SDSTS1               | ACEND      | SDIMSK1                        | ACENDM     | No                   |
|                  |                      | RSPEND     |                                | RSPENDM    |                      |
|                  | SDSTS2               | ILA        | SDIMSK2                        | ILAM       |                      |
|                  |                      | BWE        |                                | BWEM       |                      |
|                  |                      | BRE        |                                | BREM       |                      |
|                  |                      | RSPTO      |                                | RSPTOM     |                      |
|                  |                      | ILR        |                                | ILRM       |                      |
|                  |                      | ILW        |                                | ILWM       |                      |
|                  |                      | DTO        |                                | DTTOM      |                      |
|                  |                      | ENDE       |                                | ENDEM      |                      |
|                  |                      | CRCE       |                                | CRCEM      |                      |
|                  |                      | CMDE       |                                | CMDEM      |                      |
|                  |                      |            |                                |            |                      |
|                  |                      |            |                                |            |                      |
|                  |                      |            |                                |            |                      |
| SDACI            | SDIOSTS              | EXWT       | SDIOIMSK                       | EXWTM      | No                   |
|                  |                      | EXPUB52    |                                | EXPUB52M   |                      |
|                  |                      | IOIRQ      |                                | IOIRQM     |                      |
| CDETI            | SDSTS1               | SDD3IN     | SDIMSK1                        | SDD3INM    | No                   |
|                  |                      | SDD3RM     |                                | SDD3RMM    |                      |
|                  |                      | SDCDIN     |                                | SDCDINM    |                      |
|                  |                      | SDCDRM     |                                | SDCDRMM    |                      |
| SBFAI            | SDSTS2               | BWE        | SDDMAEN                        | DMAEN      | Yes                  |
|                  |                      | BRE        |                                |            |                      |

#### 48.4.1 DMA Transfer Triggered by Interrupt Requests

When the SBFAI interrupt is requested, DMA/DTC transfer can be used to write to or read the SDBUFR register. When using the SBFAI interrupt, set the SDDMAEN.DMAEN bit to 1, the SDIMSK2.BWEM bit to 1, and SDIMSK2.BREM bit to 1.

When the SDDMAEN.DMAEN bit is 1, if a write command is issued, the SDSTS2.BWE flag becomes 1; if a read command is issued, the SDSTS2.BRE flag becomes 1. At this point, the SBFAI interrupt request is output. When the last data of a block is transferred (one block is the transfer data size set in the SDSIZE.LEN[9:0] bits), the SBFAI interrupt request is cleared, and the SDSTS2.BWE flag or the SDSTS2.BRE flag becomes 0.

The SBFAI interrupt request is also cleared by following:

- The SDRST.SDRST bit is set to 0 (SDHI software reset).
- The SDSTOP.STP bit is set to 1.
- The SDIOMD.IOABT bit is set to 1.
- The SDDMAEN.DMAEN bit is set to 0.

However, if the DMAEN bit is set to 1 again before the next command is written to the SDCMD register, the SBFAI interrupt request is output again.

The SBFAI interrupt request will not be cleared when a communication error or timeout occurs during DMA transfer. Perform error processing by software.

The SDSTS2.BWE flag and BRE flag will not become 0 when a communication error or timeout occurs, nor will they become 0 when the SDSTOP.STP bit and SDIOMD.IOABT bit are set to 1. If the SDSTS2.BWE flag or BRE flag remain set to 1, even if a write command or read command is issued, the SBFAI interrupt request will not be output, and the SDSTS2.BWE flag and BRE flag must be set to 0 before issuing the next command.

Table 48.9 lists the DMAC and DTC settings when performing DMA transfer.

**Table 48.9 DMAC and DTC Settings When Performing DMA Transfer**

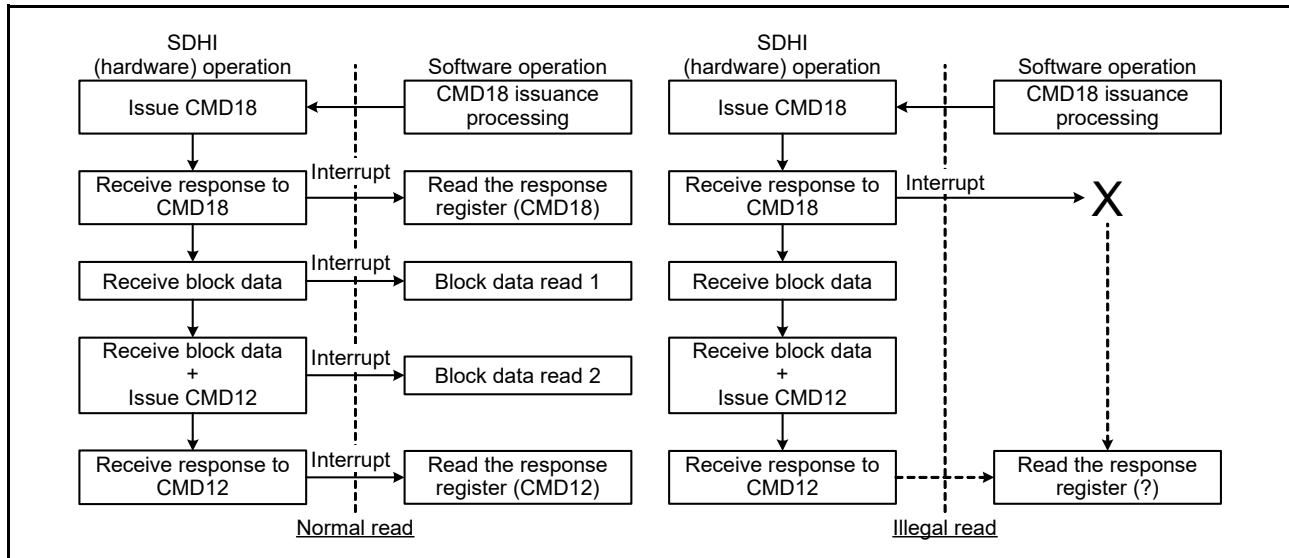
| Item                      |            | Setting Description                               |
|---------------------------|------------|---------------------------------------------------|
| Transfer mode             |            | Block transfer mode                               |
| Transfer data             | 1 data     | 32 bits                                           |
|                           | Block size | Size set in the SDSIZE.LEN[9:0] bits divided by 4 |
| Number of block transfers |            | Number of transfers set in the SDBLKCNT register  |

## 48.5 Notes on Using the SDHI

### 48.5.1 Illegal Read Access During a Multi-Block Read and How To Avoid It

When the multi-block read command (CMD18) is issued to read one or two blocks, if the response to CMD18 stored in the SDRSP10 register is read, the timing of the read access may cause the response to be read incorrectly.

Figure 48.23 shows examples of a normal read and an illegal read when using CMD18 to read two blocks.



**Figure 48.23 Multi-Block Read Processing When Reading Two Blocks**

In the example of the illegal read, when the interrupt is generated by receiving the response to CMD18, the timing for reading the SDRSP10 register during the interrupt handling is delayed, and instead of the response for CMD18 being read, the response for CMD12 or the data during the response for CMD12 may be read.

This problem can be avoided by performing either of the following:

- When reading one block or two blocks, use a single block read command instead of a multi-block read command.
- When reading the response for CMD18, read the SDRSP54 register instead of the SDRSP10 register.

When using a multi-block read command to read three or more blocks, CMD12 will not be issued if no block data is read, which will prevent this problem from happening. Also, when using a multi-block write command, the above problem can be avoided by transmitting block data after reading the response to CMD25.

### 48.5.2 SDBUFR Register Illegal Write Error

When writing to the SDBUFR register after issuing a single block write command or a multi-block write command, write data for the size set by the SDSIZE.LEN[9:0] bits. If the amount of data written to the SDBUFR register is greater than the size set by the SDSIZE.LEN[9:0] bits, an illegal write error occurs in the SDBUFR register, and the SDSTS2.ILW flag becomes 1. However, the padding data included in the data being written to the SDBUFR register is ignored, so this error does not occur. For example, if the data size set by the SDSIZE.LEN[9:0] bits is an odd number, of the data written to the SDBUFR register, there is a remainder of 1 byte or 3 bytes. Although the extra data is written to the register, no error occurs. If the data size set by the SDSIZE.LEN[9:0] bits is an even number, and there is a remainder of 2 bytes, no error occurs even if these 2 bytes are written to the SDBUFR register.

If data written to the SDBUFR register is not transmitted, the SDSTS2.SDCLKCREN flag may remain set to 0. In this case, in order to set the SDSTS2.SDCLKCREN flag to 1, the SDRST.SDRST bit must be set to 0 and then set back to 1.

### 48.5.3 Automatic Control of the SDHI Clock Output

As per the SD card specifications, after MCU power-on, 74 cycles of the SDHI clock must be output from the host to the SD card before the card initialization command (CMD0) can be issued. Therefore, 74 cycles of the SDHI clock should be output from the SDHI to the SD card before enabling automatic control of the SDHI clock output.

When automatic control of the SDHI clock output is enabled, SDHI clock output stops if the command sequence is ended by a communication error or timeout. Therefore, if it is necessary to change the internal status of the SD card even after the command sequence ends, disable automatic control of the SDHI clock output and output the SDHI clock to the SD card.

### 48.5.4 Restrictions on Setting the C52PUB Bit During a Multi-Block Write Sequence

During a CMD53 multi-block write sequence, if the SDIOMD.C52PUB bit is set to 1, the SDHI issues CMD52 after the SD buffer becomes empty. To immediately issue CMD52, perform one of the procedures below to suspend writing to the SD buffer, and set the C52PUB bit to 1.

#### Procedure to suspend writing to the SD buffer when not performing DMA transfer (interrupt used)

1. Set the SDIMSK2.BWEM bit to 1 to disable the interrupt, and suspend writing to the SDBUFR register.
2. Set the SDIOMD.C52PUB bit to 1. Then, when the SD buffer becomes empty, the SDHI issues CMD52.
3. After receiving the response for CMD52, set the SDIMSK2.BWEM bit to 0 to enable the interrupt, and resume writing to the SDBUFR register.

#### Procedure to suspend writing to the SD buffer when performing DMA transfer

1. Configure settings to perform DMA transfer every [SDSIZE register setting value  $\times n$  blocks], and suspend writing to the SDBUFR register before setting the SDIOMD.C52PUB bit ( $n = 1, 2, \dots$ ).
2. Set the SDIOMD.C52PUB bit to 1. Then, when the SD buffer becomes empty, the SDHI issues CMD52.
3. After receiving the response for CMD52, resume DMA transfer to the SDBUFR register.

### 48.5.5 Note on Setting the SDCLKCR Register

The SDCLKCR register cannot be written when the SDSTS2.SDCLKCREN flag is 0. Set the SDSTS2.SDCLKCREN flag to 1 before writing to the SDCLKCR register.

### 48.5.6 Writing to the SDSTOP Register During a Multi-Block Read Sequence

When the SDSTOP.SDBLKCNTEN bit is 1 during a multi-block read sequence, if the SDSTOP.STP bit is set to 1 and the command sequence is stopped, the command sequence may not be completed depending on when the SDSTOP.STP bit is set to 1. To avoid this problem, set the SDSTOP.STP bit and the SDSTOP.SDBLKCNTEN bit to 0 simultaneously. Note that at this time, the SDSTOP.SDBLKCNTEN bit should be set to 0 even if the SDSTS2.SDCLKCREN bit is 0.

If the SDSTOP.SDBLKCNTEN bit is not set to 0, the command sequence can be completed by setting the SDRST.SDRST bit to 0.

During a CMD53 multi-block transfer, when stopping data transfer by setting the SDIOMD.IOABT bit to 1, the SDSTOP.SDBLKCNTEN bit should remain set to 1.

### 48.5.7 Controlling Module Operation

SDHI operation is controlled by setting the MSTPCRD.MSTPD19 bit. Setting the MSTPD19 bit to 0 enables the SDHI; setting the MSTPD19 bit to 1 disables the SDHI. The SDHI is disabled after a reset. Registers in the SDHI can be accessed by setting the MSTPD19 bit to 0. Refer to section 11, Low Power Consumption for details.